How-To Library

Informatica

Implementing Oracle to Snowflake Synchronization Using Cloud Mass Ingestion Databases

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Abstract

This article shows how to set up a Mass Ingestion Databases job that performs an initial load of Oracle data to a Snowflake Data Cloud target and then replicates the incremental CDC changes to the target.

Supported Versions

- Informatica Intelligent Cloud Services November 2023

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Overview

This article walks through the steps required to initially set up a Mass Ingestion Databases environment and then configure and implement a database ingestion combined initial and incremental load job that replicates Oracle data to

a Snowflake Data Cloud target. The initial load portion of the job initially populates the Snowflake target with point-in-time data from Oracle. The incremental load portion of the job replicates the Oracle change data in real time to keep the source and target in sync. In this scenario we will also set schema drift options to automatically replicate some types of source DDL changes to the target.

The use cases for replicating real-time change data from a mission-critical Oracle production system to a Snowflake data warehouse include:

- Minimizing intrusive, non-critical work, such as offline reporting or analytical operations on the Oracle source database system.
- Reducing costly CPU usage, disk space, and CDC processing time on the Oracle source database system.
- Getting the latest database changes to your data warehouse in real time for downstream processing. After an initial batch load of data to the data warehouse, Mass Ingestion Databases can propagate data changes continuously from a source database to keep the data in the data warehouse up to date.
- Migrating to a subscription-based Cloud environment.

For more information, see the Cloud Mass Ingestion documentation on the Doc Portal at https://docs.informatica.com/integration-cloud/cloud-mass-ingestion/current-version.html.

Task flow

Perform the following tasks, in the order shown, to implement Oracle to Snowflake combined initial and incremental load processing.

1. "Initial setup of your environment" on page 3.
2. "Creating projects and project folders" on page 10 in which to store your database ingestion tasks.
3. Prepare and learn about "Oracle sources" on page 11 and "Snowflake targets" on page 24.
4. "Configuring connections" on page 25 if not already available.
5. "Configuring a database ingestion task" on page 36.
6. "Deploying the database ingestion task" on page 47.
7. "Running the deployed database ingestion job" on page 47.
8. "Monitoring database ingestion jobs" on page 47.

Initial setup of your environment

You must complete some initial setup tasks in your Informatica Intelligent Cloud Services environment before you start configuring connections and database ingestion tasks.

1. Verify that your source and target versions are supported.
2. Check that you system meets the minimum system requirements.
3. Create an organization and add users to it.
4. Download and install a Secure Agent.
5. Configure the Database Ingestion agent service for your runtime environment.

Step 1: Verify that your source and target versions are supported

To determine the source and target types and versions that are supported for each ingestion type, check the KB article What are the supported sources and targets for IICS Cloud Mass Ingestion service?.

To determine the web browsers that are compatible with Mass Ingestion and the operating systems that are supported for the Secure Agent, check the Product Availability Matrix (PAM) for Informatica Intelligent Cloud Services (IICS) on

the Informatica Network at
https://network.informatica.com/community/informatica-network/product-availability-matrices/.

Step 2: Check Mass Ingestion Databases system requirements

The following table lists the Mass Ingestion Databases minimum system requirements for the Secure Agent:

Component	Minimum requirement
Cores per CPU	8 minimum, 16 recommended if you need to process a large number of source tables in an initial load
Memory	32 GB
Disk space	5 GB per job, based on a row size of 2 KB

Step 3: Set up an organization and organization users

An organization is a secure area within the Informatica Intelligent Cloud Services repository that stores your licenses, user accounts, ingestion tasks, and information about jobs and security. Set up an organization in Administrator.

If your organization has the Organization Hierarchy license, you can also create one or more sub-organizations within your organization. You can create sub-organizations to represent different business environments in your company. For example, you might create separate sub-organizations to represent your development, testing, and production environments.

Perform the following tasks:

- On the **Organization** page, configure organization properties such as the organization name and address, authentication information, and notification email addresses. Click the Sub-organizations tab to add a sub-organization to the organization.
- On the **Users** and **User Groups** pages, add and configure users and user groups for the organization. On the User Roles page, you can configure user role permissions, which can be assigned to users and user groups.

Administrator provides context-sensitive help for all of these pages. To display the help for the page in which you're working, click the help (?) icon and select **Online Help**.

Step 4: Download and install a Secure Agent

A Secure Agent is a lightweight program that runs tasks and enables secure communication across the firewall between your organization and Informatica Intelligent Cloud Services.

Install the Secure Agent on all the systems from which you want to use Mass Ingestion Databases. You can install a Secure Agent on Windows or Linux.

When you download and install a Secure Agent, a *Secure Agent group*, also called a runtime environment, is created. A Secure Agent group can contain one Secure Agent or multiple agents.

Note: Mass Ingestion Databases does not support the Hosted Agent or serverless runtime environment.

Downloading and installing the Secure Agent on Linux

To install the Secure Agent on a Linux machine, download and run the Secure Agent installer and then register the agent.

Before you download and install the Secure Agent, verify that no other Secure Agent is installed on the machine using the same Linux user account. If any other Secure Agent exists, you must uninstall it first.

Tip: To verify the checksum of the Secure Agent installation program, use the agent REST API version 2 resource. For more information about the agent resource, see *REST API Reference*.

1. Open Administrator and select **Runtime Environments**.
2. On the **Runtime Environments** page, click **Download Runtime Installer**.
3. Select **Secure Agent** as the Environment Type.
4. Select **Linux 64** as the Platform.
5. Click **Copy** to copy the install token.

   You will need this token to install the Secure Agent. The token expires after 24 hours. If the token expires before you install the Secure Agent, repeat steps 1 to 5 to obtain a new token. You don't need to download the installer again.
6. Click **Download**.

   The installer is downloaded to your machine. The name of the installer is `agent64_install_ng_ext.<agent core version>.bin`.
7. Save the installer to a directory on the machine where you want to run the Secure Agent.

   Note: Ensure that the file path doesn't contain spaces or multibyte characters. If the file path contains spaces, the installation might fail. If the path contains multibyte characters, the Secure Agent might not start.
8. From a shell command line, navigate to the directory where you downloaded the installation program and enter the following command:

   ```
   ./agent64_install_ng_ext.bin -i console
   ```
9. When the installer completes, navigate to the following directory:

   ```
   <Secure Agent installation directory>/apps/agentcore
   ```
10. To start the Secure Agent, enter the following command:

    ```
    ./infaagent startup
    ```

    The Secure Agent Manager starts. Register the agent using the user name that you use to access Informatica Intelligent Cloud Services and also supply the install token.
11. To register the agent, in the `<Secure Agent installation directory>/apps/agentcore` directory, enter one of the following commands using your Informatica Intelligent Cloud Services user name and the token that you copied:
    - To add the agent to its own Secure Agent group, use the following command:

      ```
      ./consoleAgentManager.sh configureToken <user name> <install token>
      ```
    - To add the agent to an existing Secure Agent group, use the following command:

      ```
      ./consoleAgentManager.sh configureTokenWithRuntime <user name> <install token>
      <Secure Agent group name>
      ```

      Note: If the command includes a Secure Agent group name that doesn't exist, the Secure Agent is not assigned to a group. Be sure to use a valid Secure Agent group name.

The following table lists the command options:

Option	Description
User Name	Required. Informatica Intelligent Cloud Services user name of the user installing the Secure Agent.
Install Token	Required. The install token that you copied when you downloaded the runtime installer.
Secure Agent group name	Optional. Include a group name when you want to add the agent to an existing Secure Agent group instead. If this option isn't included in the command, the agent will be in its own Secure Agent group.

You can check the registration status of a Secure Agent using the following command:

```
./consoleAgentManager.sh isConfigured
```

Downloading and installing the Secure Agent on Windows

To install the Secure Agent on a Windows machine, you must download and run the Secure Agent installation program and then register the agent.

Secure Agent registration requires an install token. To get the install token, copy the token when you download the agent or use the **Generate Install Token** option in Administrator. The token expires after 24 hours.

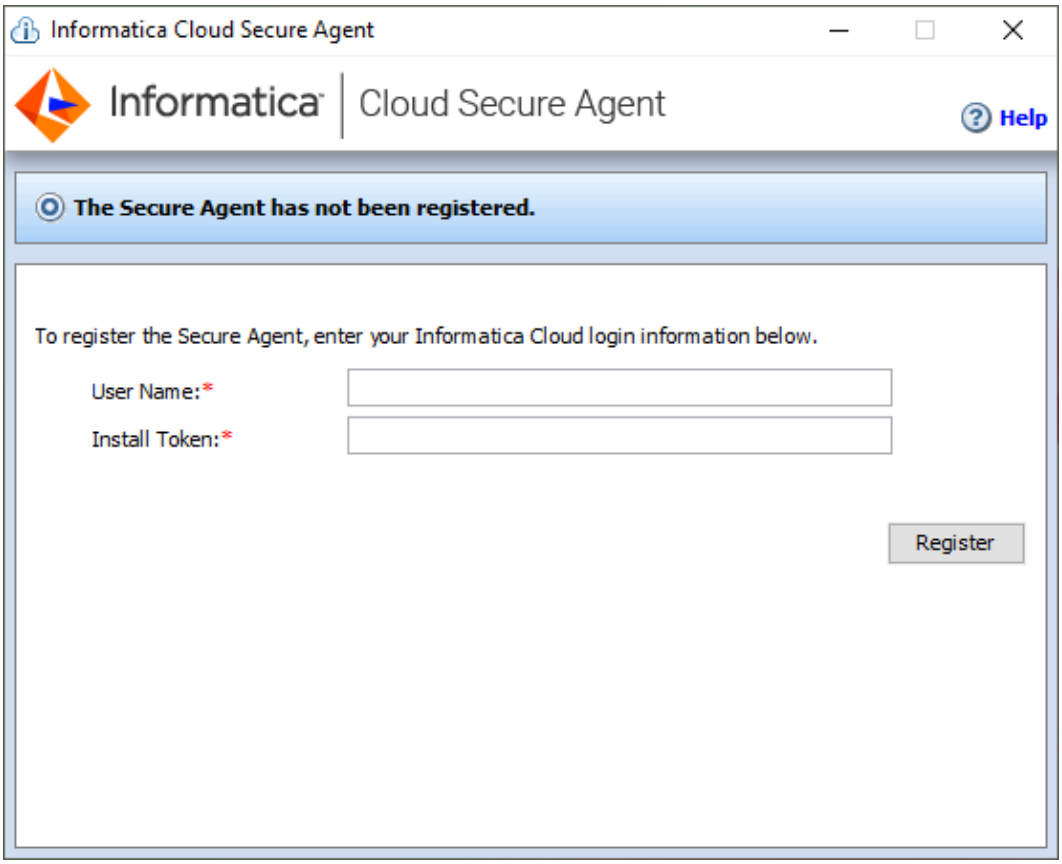
Before you download and install the Secure Agent, verify that no other Secure Agent is installed on the machine. If there is, you must uninstall it.

Tip: To verify the checksum of the Secure Agent installation program, use the agent REST API version 2 resource. For more information about the agent resource, see *REST API Reference*.

1. Open Administrator and select **Runtime Environments**.
2. On the **Runtime Environments** page, click **Download Secure Agent**.
3. Select the Windows 64-bit operating system platform, copy the install token, and then click **Download**.

   The installation program is downloaded to your machine. The name of the installation program is `agent64_install_ng_ext.<agent core version>.exe`.
4. Run the installation program as an Administrator:
   a. Specify the Secure Agent installation directory, and click **Next**.
   b. Click **Install** to install the agent.

The Secure Agent Manager opens and prompts you to register the agent as shown in the following image:



5. If you did not copy the install token when you downloaded the agent, click **Generate Install Token** on the **Runtime Environments** page in Administrator, and copy the token.
6. In the Secure Agent Manager, enter the following information, and then click **Register**:

Option	Description
User Name	User name that you use to access Informatica Intelligent Cloud Services.
Install Token	Token that you copied.

The Secure Agent Manager displays the status of the Secure Agent. It takes a minute for all of the services to start.

7. If your organization uses an outgoing proxy server to connect to the internet, enter the proxy server information.
8. Close the Secure Agent Manager.

The Secure Agent Manager minimizes to the taskbar and continues to run as a service until stopped.

Step 5: Configure the Database Ingestion service for your runtime environment

Mass Ingestion Databases uses the Database Ingestion agent service to run database ingestion jobs. The Database Ingestion service is automatically downloaded to the system when you download the Secure Agent.

After you download the Secure Agent to your runtime environment and enable the Database Ingestion service, the Database Ingestion packages are pushed to the on-premises system where the Secure Agent runs. You can then optionally configure properties for the Database Ingestion service that runs on the Secure Agent.

Database Ingestion service properties

To change or optimize the behavior of the Database Ingestion service that your Secure Agent group uses, configure Database Ingestion properties for your runtime environment.

To configure the properties, open your runtime environment and click **Edit**. Under **System Configuration Details**, select the **Database Ingestion** service and the **DBMI_AGENT_CONFIG** type.

The following table describes the Database Ingestion agent service properties:

Property	Description
maxTaskUnits	The maximum number of database ingestion task units that can run concurrently on an on-premises machine where the Secure Agent is running. Task units are not related to the capacity and availability of your hardware or software. You can configure maxTaskUnits to precisely control CPU usage. Valid values are 1 to 2000000000 (2 billion). To calculate a reasonable number of task units for your Secure Agent machine, Informatica recommends that you divide the number of cores by 3 or 4. For example, if you have an 8-core machine, you could set this property to 2. Then monitor CPU usage and adjust the property value as needed to tune performance. During initial load processing, this property determines the number of tables that can be unloaded simultaneously. Remaining tables are queued and start unload processing when resources become available. **Note:** A single job can process many tables. The total number of tables that can be processed is limited only by available memory. On the average, 25 MB of RAM is required per table for an initial load task based on a 1 KB row size. During incremental load processing, this property determines the number of database ingestion jobs that can run simultaneously. Setting this property to a value greater than the number of cores on the Secure Agent machine can increase parallelism for task execution but also cause performance bottlenecks at task execution time.
serviceLogRetentionPeriod	The number of days to retain each internal Database Ingestion service log file after the last update is written to the file. When this retention period elapses, the log file is deleted. The default value is 7 days. Service logs are retained on the Secure Agent host where they are created: `<infaagent>/apps/Database_Ingestion/logs`.

Property	Description
taskLogRetentionPeriod	The number of days to retain each job log file after the last update is written to the file. When this retention period elapses, the log file is deleted. The default value is 7 days.
ociPath	For Oracle sources and targets, the path to the Oracle Call Interface (OCI) directory that contains the oci.dll or libclntsh.so file. By default, Oracle uses $ORACLE_HOME/lib on Linux or %ORACLE_HOME%\bin on Windows. The OCI library is used by database ingestion CDC tasks to connect to Oracle. For a DBMI agent that is running, this value is appended to the PATH environment variable value on Windows or to the LD_LIBRARY_PATH environment variable value on Linux. This property is not required if you already included the OCI path in the PATH or LD_LIBRARY_PATH environment variable.
serviceUrl	The URL that the Database Ingestion service uses to connect to the Informatica Intelligent Cloud Services cloud.
logLevel	The level of detail to include in the logs that the Database Ingestion service produces. Options are: - TRACE - DEBUG - INFO - WARN - ERROR The default value is TRACE.
taskExecutionHeapSize	The maximum heap size, in gigabytes, for the Task Execution service. This value, in conjunction with maxTaskUnits property, affects the number of concurrent database ingestion tasks that can run on a Secure Agent. Try increasing the heap size to run more tasks concurrently. Enter this value followed by "g" for gigabytes, for example, '9g'. The default value is '8g'.
useProxy	Set this property to true to enable the DBMI Agent to go through a proxy when connecting to or writing data to targets. The DBMI Agent then uses the proxy settings from the Secure Agent proxy configuration. By default, proxy settings are not used.
intermediateStorageDirectory	For incremental load and combined initial and incremental load jobs, the local root directory under which intermediate files that contain data are stored when the **Enable Persistent Storage** option is selected in the associated task definitions.
storageBackupDirectory	For incremental load and combined initial and incremental load jobs, the path to the directory that stores backup files when the **Enable Persistent Storage** option is selected in the associated task definitions.

Property	Description
storageProperties	For incremental load and combined initial and incremental load jobs, a comma-separated list of key=value pairs that is used when the **Enable Persistent Storage** option is selected in the associated task definitions. Specify this property only at the direction of Informatica Global Customer Support.
task_container.jvm.allowExceptionForInvalidEncodedData	If you receive transliteration errors that report invalid encoding to UTF-8, and you do not want to repair or correct the source data, set this property to false so that database ingestion jobs do not fail when trying to unload the data from the source. With this setting, the Database Ingestion service passes an equivalent Java property to the DataDirect JDBC driver to prevent the exception from occurring. After you set this property, you must restart the Database Ingestion service.

Database Ingestion service environment variable

To change or optimize the behavior of the Database Ingestion Agent, define the following environment variable:

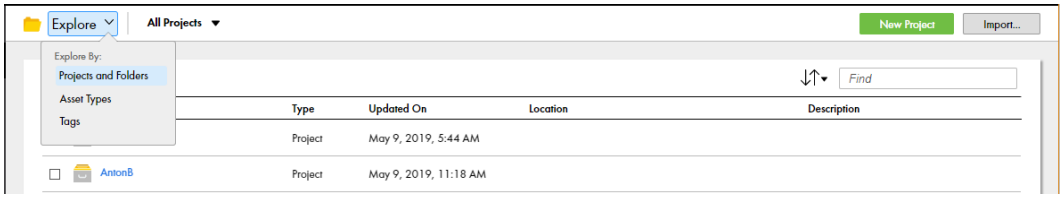
Environment Variable	Description
DBMI_WRITER_CONN_POOL_SIZE	Indicates the number of connections that a database ingestion job uses to propagate the change data to the target. The default value is 8. Valid values are 4 through 8.

Note: After you define or change an environment variable, restart the Database Ingestion Agent for the changes to take effect.

Creating projects and project folders

Create a project to contain your ingestion task assets. You can create one or more folders under the project to logically organize your assets. However, you cannot create subfolders under a folder.

1. In the Mass Ingestion service, open the **Explore** page.
2. If the **Explore** page shows objects other than projects, select **Projects and Folders** in the **Explore** menu.



3. To create a project, click **New Project**.
4. In the **New Project Properties** dialog box, enter a project name up to 255 characters in length. You can also enter an optional description of the project.

   A project name cannot contain the following characters: `#?`|{}"^&[]/\.`
5. If you want to add a folder under the project, select the project and click **New Folder**.

6. In the **New Folder Properties** dialog box, enter a folder name up to 255 characters in length. You can also enter an optional description of the project.

   A folder name cannot contain the following characters: `#?`|{}"^&[]/\.`

When you define an ingestion task, you must specify the project or project folder location to contain the task definition.

Oracle sources

To use Oracle sources in database ingestion tasks, first prepare the source database and review the usage considerations.

Source preparation

- Verify that you use an Oracle version that Mass Ingestion Databases supports. See the KB article FAQ: What are the supported sources and targets for IICS Cloud Mass Ingestion service?.
- Define the following system environment variables on the Linux or Windows system where the Secure Agent runs:
  - ORACLE_HOME environment variable. Points to the Oracle client installation directory, one level up from the bin directory on Windows or lib directory on Linux. This environment variable is not required. However, if you do not define it, you'll need to specify the full Oracle client installation path when you define other environment variables or Secure Agent properties that specify a path to a subdirectory.
  - TNS_ADMIN environment variable. If you specify a TNS name in the **Database Connect String** property of the Oracle Database Ingestion connection properties, use this environment variable to point to the directory location of the tsnnames.ora file when the file is not in the default $ORACLE_HOME/network/admin directory. The tnsnames.ora file, along with the Oracle Call Interface (OCI), is used to communicate with the Oracle source database.

  Note: In Administrator, you can set the ociPath property for the Database Ingestion agent service (DBMI agent) to point to the OCI library that contains the oci.dll or libclntsh.so file. The OCI library is used by database ingestion CDC tasks to connect to Oracle. By default, Oracle uses the ociPath value of $ORACLE_HOME/lib on Linux or %ORACLE_HOME%\bin on Windows.
- Make sure the Mass Ingestion Databases user has the Oracle privileges that are required for the database ingestion load type to be performed.

  Note: For combined initial and incremental loads, make sure that the GRANT FLASHBACK privilege is issued for each selected source table or use the ANY TABLE option. Mass Ingestion Databases uses an Oracle Flashback Query, which consists of a SELECT AS OF *scn* statement, to query for row data from source tables in the Oracle database. Oracle requires the GRANT FLASHBACK privilege for this query to be used.

  For more information, see "Oracle privileges" on page 14.
- Database ingestion jobs require read access to Oracle online and archive redo logs to read incremental change data. If the redo logs are remote from the on-premises system where the Secure Agent runs, make sure that read access to the logs is provided, for example, by using Oracle Automatic Storage Management (ASM), mounting the logs to a network file system (NFS), or configuring BFILE access to logs that are on the Oracle file system.
- If you plan to read data from redo log files in Oracle ASM, Informatica recommends that you set the sqlnet.recv_timeout parameter in the local sqlnet.ora file to less than 5 minutes. This parameter specifies the duration of time that the Oracle client waits for a response from ASM before a query times out. Network interrupts and other factors can occasionally make Oracle connections unresponsive. Setting this value ensures that the reader can respond and recover from any such situation in a timely manner.
- If you use Oracle 11.2.04, set the Oracle COMPATIBLE initialization parameter to 11.2.04 to ensure that Oracle has all of the most current Redo Logs fixes for that release.

- Ensure that the Oracle Database Client or Instant Client is installed and configured on the Secure Agent server for the Secure Agent to communicate with Oracle. If you do not already have an Oracle client installed, you can download a client and access installation information from the Oracle web site, or ask your Oracle DBA to download and configure an Oracle client.
- For incremental load or combined initial and incremental load operations, perform the following prerequisite tasks in Oracle:
  - Enable ARCHIVELOG mode for the Oracle database. If the database is not in an Amazon RDS environment, issue the following SQL statements:

    ```
    SHUTDOWN IMMEDIATE;
    STARTUP MOUNT;
    ALTER DATABASE ARCHIVELOG;
    ALTER DATABASE OPEN;
    SHUTDOWN IMMEDIATE;
    STARTUP;
    ```

    For an Amazon RDS for Oracle databases, set the backup retention period to place the database in ARCHIVELOG mode and enable automated backups.
  - Define an archive log destination.
  - Enable Oracle minimal global supplemental logging on the source database.
  - If your Oracle source tables have primary keys, ensure that supplemental logging is enabled for all primary key columns. For source tables that do not have primary keys, ensure that supplemental logging is enabled for all columns from which change data will be captured.

    Note: When you create a database ingestion task, you have the option of generating a script that implements supplemental logging for all columns or only primary key columns for the selected source tables.
  - Ensure that the Oracle MAX_STRING_SIZE initialization parameter is *not* set to EXTENDED. If it is set to EXTENDED, Mass Ingestion Databases will not be able to replicate inserts and updates for tables containing columns defined with large (extended size) VARCHAR2, NVARCHAR2, or RAW columns.

  If you do not have the authority to perform these tasks, ask your Oracle database administrator to perform them. For more information, see the Oracle documentation.

Amazon Relational Database Service (RDS) for Oracle source preparation:

1. Create the ONLINELOG_DIR and ARCHIVELOG_DIR directories that will hold the online and archive redo logs, respectively, on the RDS file system. Use the following exec statements:

   ```
   exec rdsadmin.rdsadmin_master_util.create_archivelog_dir;
   exec rdsadmin.rdsadmin_master_util.create_onlinelog_dir;
   ```
2. Grant the Oracle privileges that are required for the Amazon RDS for Oracle source type to the Mass Ingestion Databases user.
   For more information about the privileges required for an Amazon RDS for Oracle source, see "Oracle privileges for Amazon RDS for Oracle sources" on page 16.
3. Define an appropriate retention time for the archived redo logs. Use the following exec statement:

   ```
   exec rdsadmin.rdsadmin_util.set_configuration('archivelog retention
   days',number_of_days);
   ```
4. In the Amazon RDS console, set the backup retention period for the source database to a value greater than zero to enable automated backups of the database instance.

   Note: This step enables ARCHIVELOG mode for the database.
5. Ensure that supplemental logging is enabled at the database level. Use the following statement:

   ```
   exec rdsadmin.rdsadmin_util.alter_supplemental_logging('ADD');
   ```

   When you create a database ingestion task, you can generate a script to enable supplemental logging for the selected source tables.

6. Optionally, in the Amazon RDS console, you can create a parameter group and define the cache sizes of the default buffer pool. The default buffer pool holds buffers that use the primary database block size. Use the following DB_CACHE_SIZE parameter values:
   - DB_2K_CACHE_SIZE
   - DB_4K_CACHE_SIZE
   - DB_16K_CACHE_SIZE
   - DB_32K_CACHE_SIZE

   Then select the parameter group for the source database.

Usage considerations

- Because Mass Ingestion Databases expects each source table row to be unique, Informatica recommends that each source table have a primary key. Mass Ingestion Databases does not honor unique indexes in place of a primary key. If no primary key is specified, Mass Ingestion Databases treats all columns as if they are part of the primary key.
- For Oracle sources that use the multitenant architecture, the source tables must reside in a single pluggable database (PDB) within a multitenant container database (CDB).
- You can use Oracle Transparent Data Encryption (TDE) to encrypt data in tablespaces that contain Oracle source tables for incremental load processing. Mass Ingestion Databases supports storing the master encryption key in a TDE keystore that is in a file system, in ASM, or in an external hardware security module (HSM) that supply PKCS11 interfaces, such as Oracle Key Vault (OKV). For more information, contact Informatica Global Customer Support.
- Mass Ingestion Databases can process data across a RESETLOGS boundary. To avoid the source and targets becoming out of sync, Informatica recommends that you stop capture processing before performing a RESETLOGS and then restart capture processing after the RESETLOGS event. Otherwise, the capture process might send data to the target that is subsequently reverted by the RESETLOGS event, causing the source and target to become out of sync.
- Alternative strategies for accessing the Oracle redo logs are available. For more information, see "Oracle log access methods for CDC" on page 19.
- If a database ingestion incremental load or combined initial and incremental load task contains an Oracle source table name or one or more column names that are longer than 30 characters, Oracle suppresses supplemental logging for the entire table, including primary keys and foreign keys. As a result, most operations on the table fail. This problem is caused by an Oracle restriction. In this situation, exclude the table from capture processing or rename the long table and column names to names of 30 characters or less.
- Mass Ingestion Databases does not support the following Oracle source data types:
  - "ANY types" such as ANYTYPE, ANYDATA, ANYDATASET
  - Extended types
  - INTERVAL
  - JSON
  - LOBs, except for BLOBs, CLOBs, and NCLOBs in initial load jobs
  - TIMESTAMP WITH LOCAL TIME ZONE
  - UROWID
  - XML-defined types such as XMLTYPE, URI types, URIFactory package subtypes
  - Spatial types such as SDO_GEOMETRY
  - User-defined types such as OBJECT, REF, VARRAY, nested table types

  Source columns that have unsupported data types are excluded from the target definition.

- Do not use custom data type mapping rules to map an Oracle source RAW column to a target CHAR or VARCHAR column. Otherwise, the deployment of the database ingestion task might fail.
- Mass Ingestion Databases does not support invisible columns in Oracle source columns. For these columns, database ingestion incremental load jobs and combined initial and incremental load jobs propagate nulls to the corresponding target columns.
- If you update the primary key value in the Oracle source table for a record that does not exist in the target table, the record is not replicated to the target. However, the monitoring interface increments the Update count to include the primary key update. The data will replicate to the target if the record already exists in the target table before the update of the primary key value.
- If an Update to an Oracle table does not change an existing column value, the Update count in the monitoring details for the table is still incremented but no Update row is applied to the target. Mass Ingestion Databases ignores Update rows that do not actually change values. Also, for most database targets, Mass Ingestion Databases does some aggregation of change records at a micro-batch level before writing changes to the targets. This situation can also lead to a mismatch between the Update count in the monitoring statistics and the rows applied to the target.
- The supplemental logging settings for tables might be ignored by Oracle if the table name or any table column name is longer than 30 characters. In this case, the results of database ingestion incremental load or combined load jobs are unpredictable.
- Database ingestion incremental load and combined initial and incremental load jobs that have an Oracle source and use a Secure Agent group with multiple agents can switch over to another agent in the group when the active agent stops running, subject to following limitations:
  - The job cannot have persistent storage enabled.
  - You must Stop and Resume the job for the switch to occur.

Oracle privileges

To deploy and run a database ingestion task that has an Oracle source, the source connection must specify a Mass Ingestion Databases user who has the privileges required for the ingestion load type.

Privileges for incremental load processing

For a database ingestion task that performs an incremental load or combined initial and incremental load, ensure that the Mass Ingestion Databases user (*cmid_user*) has been granted the following privileges:

Note: If the Oracle logs are managed by ASM, the user must have SYSASM or SYSDBA authority.

```
GRANT CREATE SESSION TO <cmid_user>;

GRANT SELECT ON table TO <cmid_user>;     -- For each source table created by user
GRANT EXECUTE ON DBMS_FLASHBACK TO <cmid_user>;

-- The following grant is required for combined initial and incremental loads only. Do not
-- use ANY TABLE unless your security policy allows it.
GRANT FLASHBACK ON table|ANY TABLE TO <cmid_user>;

-- Include the following grant only if you want to Execute the CDC script for enabling
-- supplemental logging from the user interface. If you manually enable supplemental
-- logging, this grant is not needed.
GRANT ALTER table|ANY TABLE TO <cmid_user>;

GRANT SELECT ON DBA_CONSTRAINTS TO <cmid_user>;
GRANT SELECT ON DBA_CONS_COLUMNS TO <cmid_user>;
GRANT SELECT ON DBA_INDEXES TO <cmid_user>;
GRANT SELECT ON DBA_LOG_GROUPS TO <cmid_user>;
GRANT SELECT ON DBA_LOG_GROUP_COLUMNS TO <cmid_user>;
GRANT SELECT ON DBA_OBJECTS TO <cmid_user>;
GRANT SELECT ON DBA_OBJECT_TABLES TO <cmid_user>;
GRANT SELECT ON DBA_TABLES TO <cmid_user>;
```

```
GRANT SELECT ON DBA_TABLESPACES TO <cmid_user>;
GRANT SELECT ON DBA_USERS TO <cmid_user>;

GRANT SELECT ON "PUBLIC".V$ARCHIVED_LOG TO <cmid_user>;
GRANT SELECT ON "PUBLIC".V$CONTAINERS TO <cmid_user>;  -- For Oracle multitenant environments
GRANT SELECT ON "PUBLIC".V$DATABASE TO <cmid_user>;
GRANT SELECT ON "PUBLIC".V$DATABASE_INCARNATION TO <cmid_user>;
GRANT SELECT ON "PUBLIC".V$ENCRYPTION_WALLET TO <cmid_user>;     -- For Oracle TDE access
GRANT SELECT ON "PUBLIC".V$LOG TO <cmid_user>;
GRANT SELECT ON "PUBLIC".V$LOGFILE   TO <cmid_user>;
GRANT SELECT ON "PUBLIC".V$PARAMETER TO <cmid_user>;
GRANT SELECT ON "PUBLIC".V$PDBS TO <cmid_user>;     -- For Oracle multitenant environments
GRANT SELECT ON "PUBLIC".V$SPPARAMETER TO <cmid_user>;
GRANT SELECT ON "PUBLIC".V$STANDBY_LOG TO <cmid_user>;
GRANT SELECT ON "PUBLIC".V$THREAD TO <cmid_user>;
GRANT SELECT ON "PUBLIC".V$TRANSACTION TO <cmid_user>;
GRANT SELECT ON "PUBLIC".V$TRANSPORTABLE_PLATFORM TO <cmid_user>;
GRANT SELECT ON "PUBLIC".V$VERSION TO <cmid_user>;

GRANT SELECT ON SYS.ATTRCOL$ TO <cmid_user>;
GRANT SELECT ON SYS.CCOL$ TO <cmid_user>;
GRANT SELECT ON SYS.CDEF$ TO <cmid_user>;
GRANT SELECT ON SYS.COL$ TO <cmid_user>;
GRANT SELECT ON SYS.COLTYPE$ TO <cmid_user>;
GRANT SELECT ON SYS.IDNSEQ$ TO <cmid_user>;
GRANT SELECT ON SYS.IND$ TO <cmid_user>;
GRANT SELECT ON SYS.INDPART$  TO <cmid_user>;
GRANT SELECT ON SYS.OBJ$ TO <cmid_user>;
GRANT SELECT ON SYS.PARTOBJ$ TO <cmid_user>;
GRANT SELECT ON SYS.RECYCLEBIN$ TO <cmid_user>;
GRANT SELECT ON SYS.TAB$ TO <cmid_user>;
GRANT SELECT ON SYS.TABCOMPART$  TO <cmid_user>;
GRANT SELECT ON SYS.TABPART$ TO <cmid_user>;
GRANT SELECT ON SYS.TABSUBPART$ TO <cmid_user>;

-- Also ensure that you have access to the following ALL_* views:
ALL_CONSTRAINTS
ALL_CONS_COLUMNS
ALL_ENCRYPTED_COLUMNS
ALL_INDEXES
ALL_IND_COLUMNS
ALL_OBJECTS
ALL_TABLES
ALL_TAB_COLS
ALL_TAB_PARTITIONS
ALL_USERS
```

Privileges for initial load processing

For a database ingestion task that performs an initial load, ensure that the user has the following privileges at minimum:

```
GRANT CREATE SESSION TO <cmid_user>;

GRANT SELECT ON DBA_INDEXES TO <cmid_user>;
GRANT SELECT ON DBA_OBJECT_TABLES TO <cmid_user>;
GRANT SELECT ON DBA_OBJECTS TO cmid_user;
GRANT SELECT ON DBA_TABLES TO <cmid_user>;
GRANT SELECT ON DBA_USERS TO <cmid_user>;
GRANT SELECT ON DBA_VIEWS TO <cmid_user>;  -- Only if you unload data from views

GRANT SELECT ANY TABLE TO <cmid_user>;
-or-
GRANT SELECT ON table TO <cmid_user>;         -- For each source table created by user

GRANT SELECT ON ALL_CONSTRAINTS TO <cmid_user>;
GRANT SELECT ON ALL_CONS_COLUMNS TO <cmid_user>;
GRANT SELECT ON ALL_ENCRYPTED_COLUMNS TO <cmid_user>;
GRANT SELECT ON ALL_IND_COLUMNS TO <cmid_user>;
GRANT SELECT ON ALL_INDEXES TO <cmid_user>;
GRANT SELECT ON ALL_OBJECTS TO <cmid_user>;
```

```
GRANT SELECT ON ALL_TAB_COLS TO <cmid_user>;
GRANT SELECT ON ALL_USERS TO <cmid_user>;

GRANT SELECT ON "PUBLIC"."V$DATABASE"  TO cmid_user;
GRANT SELECT ON "PUBLIC"."V$CONTAINERS" TO cmid_user;
GRANT SELECT ON SYS.ATTRCOL$ TO <cmid_user>;
GRANT SELECT ON SYS.CCOL$ TO <cmid_user>;
GRANT SELECT ON SYS.CDEF$ TO <cmid_user>;
GRANT SELECT ON SYS.COL$ TO <cmid_user>;
GRANT SELECT ON SYS.COLTYPE$ TO <cmid_user>;
GRANT SELECT ON SYS.IND$ TO <cmid_user>;
GRANT SELECT ON SYS.IDNSEQ$ TO cmid_user;
GRANT SELECT ON SYS.OBJ$ TO <cmid_user>;
GRANT SELECT ON SYS.RECYCLEBIN$ TO <cmid_user>;
GRANT SELECT ON SYS.TAB$ TO <cmid_user>;
```

Oracle privileges for Amazon RDS for Oracle sources

If you have an Amazon RDS for Oracle source, you must grant certain privileges to the Mass Ingestion Databases user.

Important: You must log in to Amazon RDS under the master username to run GRANT statements and procedures.

Grant the SELECT privilege, at minimum, on objects and system tables that are required for CDC processing to the Mass Ingestion Databases user (*cmid_user*). Some additional grants are required in certain situations.

Use the following GRANT statements:

```
GRANT SELECT ON "PUBLIC"."V$ARCHIVED_LOG"           TO "cmid_user";

GRANT SELECT ON "PUBLIC"."V$DATABASE"               TO "cmid_user";
GRANT SELECT ON "PUBLIC"."V$LOG"                    TO "cmid_user";
GRANT SELECT ON "PUBLIC"."V$LOGFILE"                TO "cmid_user";
GRANT SELECT ON "PUBLIC"."V$TRANSPORTABLE_PLATFORM" TO "cmid_user";
GRANT SELECT ON "PUBLIC"."V$THREAD"                 TO "cmid_user";
GRANT SELECT ON "PUBLIC"."V$DATABASE_INCARNATION"   TO "cmid_user";
GRANT SELECT ON "PUBLIC"."V$TRANSACTION"            TO "cmid_user";

GRANT SELECT ON "SYS"."DBA_CONS_COLUMNS"      TO "cmid_user";
GRANT SELECT ON "SYS"."DBA_CONSTRAINTS"       TO "cmid_user";
GRANT SELECT ON DBA_INDEXES                   TO "cmid_user";
GRANT SELECT ON "SYS"."DBA_LOG_GROUP_COLUMNS" TO "cmid_user";
GRANT SELECT ON "SYS"."DBA_TABLESPACES"       TO "cmid_user";

GRANT SELECT ON "SYS"."OBJ$"          TO "cmid_user";
GRANT SELECT ON "SYS"."TAB$"          TO "cmid_user";
GRANT SELECT ON "SYS"."IND$"          TO "cmid_user";
GRANT SELECT ON "SYS"."COL$"          TO "cmid_user";

GRANT SELECT ON "SYS"."PARTOBJ$"      TO "cmid_user";
GRANT SELECT ON "SYS"."TABPART$"      TO "cmid_user";
GRANT SELECT ON "SYS"."TABCOMPART$"   TO "cmid_user";
GRANT SELECT ON "SYS"."TABSUBPART$"   TO "cmid_user";
COMMIT;

/* For combined load jobs:*/
GRANT EXECUTE ON DBMS_FLASHBACK TO "cmid_user";

/*To provide read access to the Amazon RDS online and archived redo logs:*/
GRANT READ ON DIRECTORY ONLINELOG_DIR TO "cmid_user";
GRANT READ ON DIRECTORY ARCHIVELOG_DIR TO "cmid_user";
```

Additionally, log in as the master user and run the following Amazon RDS procedures to grant the SELECT privilege on some more objects:

```
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'DBA_TABLES',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
```

```
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'DBA_OBJECTS',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'DBA_OBJECT_TABLES',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'DBA_VIEWS',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
/begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'DBA_USERS',
p_grantee => 'cmid_user',
p_privilege => 'SELECT',
p_grant_option => false);
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'V_$CONTAINERS',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'V_$PARAMETER',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'V_$SPPARAMETER',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'V_$STANDBY_LOG',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'V_$VERSION',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'ALL_CONS_COLUMNS',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
```

```
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'ALL_CONSTRAINTS',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'ALL_OBJECTS',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'ALL_TABLES',
p_grantee => 'cmid_user',
p_privilege => 'SELECT',
p_grant_option => false);
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'ALL_TAB_PARTITIONS',
p_grantee => 'cmid_user',
p_privilege => 'SELECT',
p_grant_option => false);
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'ALL_USERS',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rd 'ALL_TABLES',
p_grantee => 'sadmin_util.grant_sys_object(
p_obj_name => 'cmid_user',
p_privilege => 'SELECT',
p_grant_option => false);
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'ALL_TAB_PARTITIONS',
p_grantee => 'cmid_user',
p_privilege => 'SELECT',
p_grant_option => false);
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'ATTRCOL$',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'CCOL$',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
```

```
p_obj_name => 'COLTYPE$',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'INDPART$',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'IDNSEQ$',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'CDEF$',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'RECYCLEBIN$',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
/* Only required for RDS21 which supports PDB*/
begin
rdsadmin.rdsadmin_util.grant_sys_object(
p_obj_name => 'V_$PDBS',
p_grantee => 'cmid_user',
p_privilege => 'SELECT');
end;
/
```

Oracle log access methods for CDC

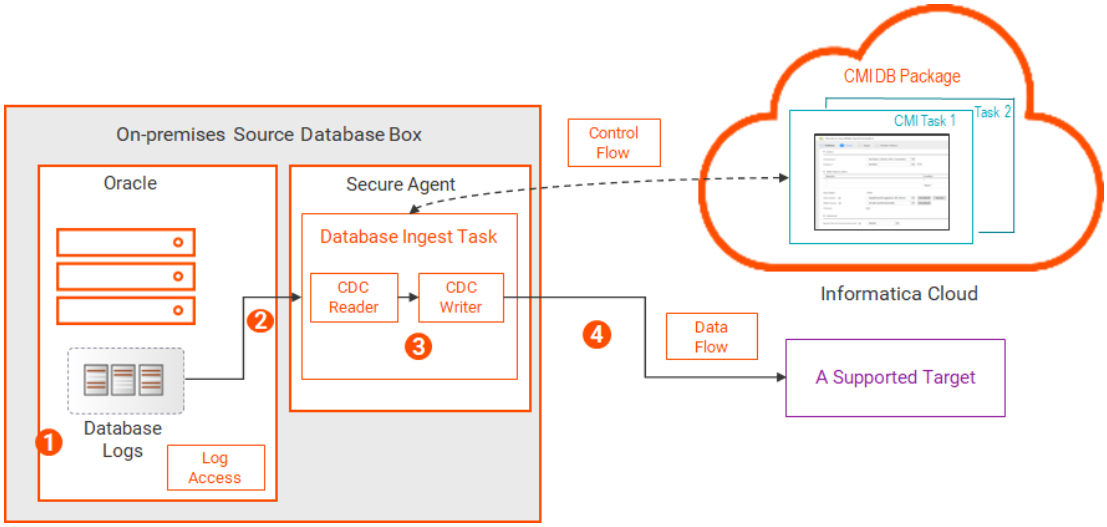
Database ingestion incremental load and combined initial and incremental load jobs can access the Oracle redo logs for CDC processing in alternative ways, depending on your environment and requirements.

Direct log access

Database ingestion jobs can directly access the physical Oracle redo logs on the on-premises source system to read change data.

Note: If you store the logs on solid-state disk (SSD), this method can provide the best performance.

The following image shows the data flow:

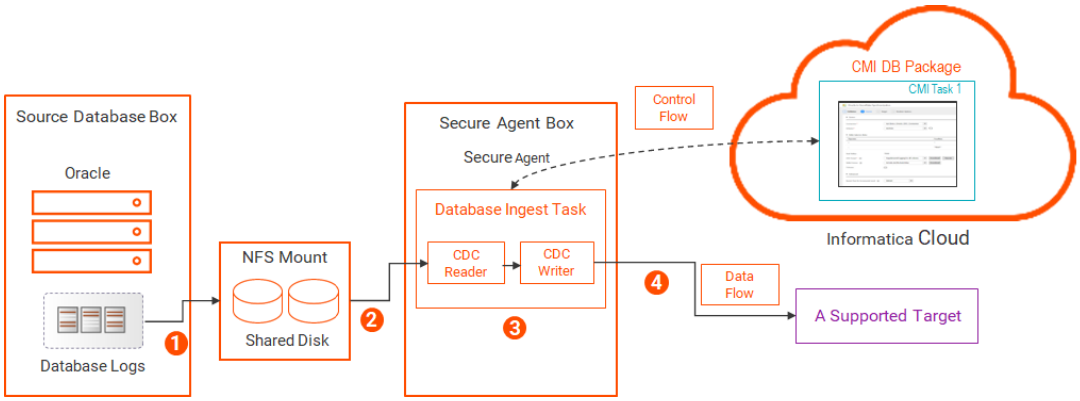


1. The Oracle database writes change records to the database log files on disk.
2. The Mass Ingestion Databases CDC Reader reads the physical log files and extracts change records from the log files for the source tables of CDC interest.
3. The Mass Ingestion Databases CDC Writer reads the change records.
4. The CDC Writer applies the change records to the target.

NFS-mounted logs

Database ingestion jobs can access to Oracle database logs from shared disk by using a Network File Sharing (NFS) mount or another method such as Network Attached Storage (NAS) or clustered storage.

The following image shows the data flow:



1. The Oracle database writes change records to database log files. The log files are written to shared disk. The shared disk can be on any system that allows the files to appear as local to both the database and Secure Agent hosts. This sharing can be achieved by using NFS, as shown above, or by using Network Attached Storage (NAS) or clustered storage.
2. The Mass Ingestion Databases CDC Reader reads the log files from the NFS server over the network and extracts the change records for the source tables of CDC interest.
3. The Mass Ingestion Databases CDC Writer reads the change records.
4. The CDC Writer applies the change records to the target.

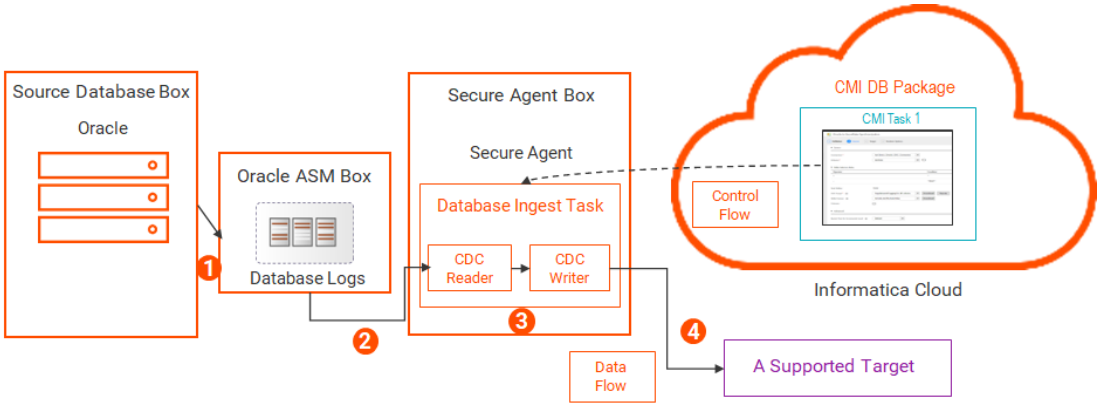
ASM-managed logs

Database ingestion jobs can access Oracle redo logs that are stored in an Oracle Automatic Storage Management (ASM) system. To read change data from the ASM-managed redo logs, the ASM user must have SYSASM or SYSDBA authority on the ASM instance.

When you configure an Oracle Database Ingestion connection, complete the properties that include "ASM" in their names.

Also, Informatica recommends that you set the sqlnet.recv_timeout parameter in the local sqlnet.ora file to less than 5 minutes when reading data from redo log files in Oracle ASM. This parameter specifies the duration of time that the Oracle client waits for a response from ASM before a query times out. Network interrupts and other factors can occasionally make Oracle connections unresponsive. Setting this value ensures that the reader can respond and recover from any such situation in a timely manner.

The following image shows the data flow:

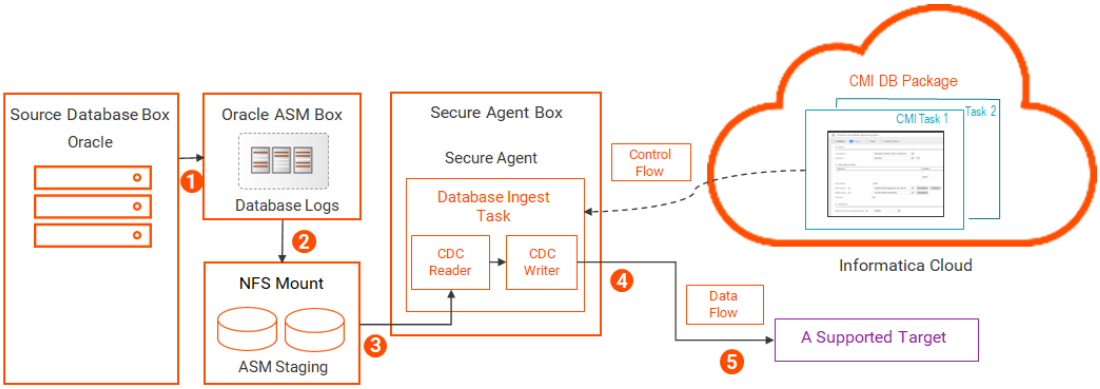


1. The Oracle database writes change records to the ASM-managed database log files.
2. The Mass Ingestion Databases CDC Reader reads the ASM-managed log files and extracts the change records for the source tables of CDC interest.
3. The Mass Ingestion Databases CDC Writer reads the change records.
4. The CDC Writer applies the change records to the target.

ASM-managed logs with a staging directory

Database ingestion jobs can access ASM-managed redo logs from a staging directory in the ASM environment. In comparison to using ASM only, this method can provide faster access to the log files and reduce I/O on the ASM system. To read change data from the ASM-managed logs, the ASM user must have SYSASM or SYSDBA authority on the ASM instance.

The following image shows the data flow:

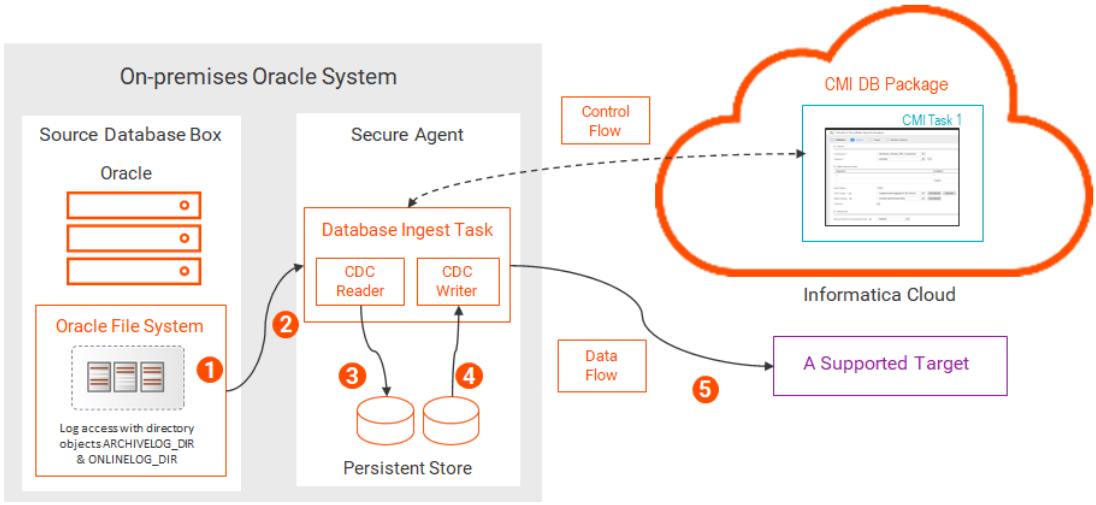


1. The Oracle database writes change records to the ASM-managed log files.
2. ASM copies the logs to a staging directory.
   The staging directory must be on shared disk, such as an NFS mount, so that ASM can write data to it and database ingestion jobs can read data from it.
3. The Mass Ingestion Databases CDC Reader reads the log files in the staging directory and extracts the change records for the source tables of CDC interest.
4. The Mass Ingestion Databases CDC Writer reads the change records.
5. The CDC Writer applies the change records to the target.

BFILE access to logs in the Oracle server file system by using directory objects

On an on-premises Oracle source system, you can configure Mass Ingestion Databases to read online and archived redo logs from the local Oracle server file system by using Oracle directory objects with BFILE locators. You must create Oracle directory objects named ARCHIVELOG_DIR and ONLINELOG_DIR that point to the locations of the Oracle redo log files. For information about configuring BFILE access, see "Configuring BFILE access to Oracle redo logs in the Oracle file system" on page 23.

The following image shows the data flow:



1. The Oracle database writes change records to the redo log files in the local Oracle server file system. When a database ingestion task needs to read log files, it connects to Oracle and issues a select request that references the ARCHIVELOG_DIR or ONLINELOG_DIR directory object to access the logs.
2. The Mass Ingestion Databases CDC Reader reads the log file and extracts the change records for the source tables of CDC interest.

3. The Mass Ingestion Databases CDC Writer reads the change records.
4. The CDC Writer applies the change records to the target.

Configuring BFILE access to Oracle redo logs in the Oracle file system

If you store redo logs in the local Oracle server file system and want to access the logs by using Oracle directory objects with BFILEs, perform the following configuration tasks:

Complete the following usual Oracle source preparation tasks, which are not specific to BFILE access:

- Define the ORACLE_HOME environment variable on the Linux or Windows system where the Secure Agent runs for Mass Ingestion Databases to use the Oracle Call Interface (OCI) to communicate with the Oracle source database.
- Make sure the Mass Ingestion Databases user has the Oracle privileges that are required for the database ingestion incremental load processing. For more information, see "Oracle privileges" on page 14.
- Enable ARCHIVELOG mode for the Oracle database.
- Define the archive log destination.

  Note: For BFILE access, use a specific archive log destination directory. Do not use the Oracle USE_DB_RECOVERY_FILE_DEST keyword to store archive logs by date in a Fast Recovery Area (FRA).
- Enable Oracle minimal global supplemental logging on the source database.
- If your Oracle source tables have primary keys, ensure that supplemental logging is enabled for all primary key columns. For source tables that do not have primary keys, ensure that supplemental logging is enabled for all columns from which change data will be captured.

  Note: When you create a database ingestion task, you have the option of generating a script that implements supplemental logging for all columns or only primary key columns for the selected source tables.
- Ensure that the Oracle MAX_STRING_SIZE initialization parameter is *not* set to EXTENDED. If it is set to EXTENDED, Mass Ingestion Databases will not be able to replicate inserts and updates for tables containing columns defined with large (extended size) VARCHAR2, NVARCHAR2, or RAW columns.

Additionally, for BFILE access, perform the following steps:

1. Query the Oracle database for the online and archived redo log locations in the Oracle server file system. You can use the following example queries:
   To get location of the online redo logs:

   ```
   select * from v$logfile;
   ```

   To get the archive log destination:

   ```
   select dest_id, dest_name, destination, status from V$ARCHIVE_DEST;
   ```
2. Create the ONLINELOG_DIR and ARCHIVELOG_DIR directory objects that point to the locations of log files from step 1. An Oracle directory object specifies a logical alias name for a physical directory on the Oracle server file system under which the log files to be accessed are located. For example:

   ```
   CREATE DIRECTORY ONLINELOG_DIR AS '/u01/oracle/data';
   CREATE DIRECTORY ARCHIVELOG_DIR AS '/u01/oracle/archivedata';
   ```

   Note: If you plan to set the reader mode to ARCHIVEONLY in the Oracle Database Ingestion connection to read changes only from archive logs, you do not need to create an ONLINELOG_DIR directory or directory object.

   The Oracle database does not verify that the directories you specify exist. Make sure you specify valid directories that exist in the Oracle file system.

3. To verify that the directory objects were created with the correct file system paths for the redo logs, issue a select statement such as:

```
select * from all_directories;
OWNER    DIRECTORY_NAME      DIRECTORY_PATH
-------- ------------------- --------------------------------
SYS      ARCHIVELOG_DIR      /u01/oracle/data/JO112DTL
SYS      ONLINELOG_DIR       /u01/oracle/data/JO112DTL
```

4. Grant read and write access on the ONLINELOG_DIR and ARCHIVELOG_DIR directory objects to the Mass Ingestion Databases user who is specified in the Oracle Database Ingestion connection properties. For example:

```
grant read on directory "ARCHIVELOG_DIR" to "cmid_user";
grant read on directory "ONLINELOG_DIR" to "cmid_user";
```

5. In the Oracle Database Ingestion connection properties, select the **BFILE Access** check box.

Snowflake targets

Target preparation

Complete the following steps as the ACCOUNTADMIN user.

1. Create a Mass Ingestion user. Use one of the following SQL statements:

```
create user INFACMI_User password  'Xxxx@xxx';
```

or

```
replace user INFACMI_User password  'Xxxx@xxx';
```

2. Create a new role and grant the role to the Mass Ingestion user. Use the following SQL statements:

```
create role INFA_CMI_Role;
grant role INFA_CMI_Role to user INFACMI_User;
```

3. Grant usage on the Snowflake virtual warehouse to the new role. Use the following SQL statement:

```
grant usage on warehouse CMIWH to role INFA_CMI_Role;
```

4. Grant usage on the Snowflake database to the new role. Use the following SQL statement:

```
grant usage, CREATE SCHEMA on database CMIDB to role INFA_CMI_Role;
```

5. Set the default role for the newly created user. Use the following SQL statement:

```
alter user INFACMI_User set default_role=INFA_CMI_Role;
```

Also, as the INFACMI_User, create a new schema:

```
create schema CMISchema;
```

Note: If the user's default role is used for ingestion tasks and does not have the required privileges, the following error will be issued at runtime:

```
SQL compilation error: Object does not exist, or operation cannot be performed.
```

Usage considerations

- Before writing data to Snowflake target tables, database ingestion jobs first write the data to an internal stage with the name you specified in the associated database ingestion task.
- When you define a connection for a Snowflake target, you must set the **Additional JDBC URL Parameters** field to `database=target_database_name`. Otherwise, when you try to define the target in the database ingestion task wizard, an error message reports that the list of schemas cannot be retrieved.
- When you define a connection for a Snowflake target using the **KeyPair** option as the authentication method and you generate the private key with OpenSSL 3.x.x version, use `PBE-SHA1-2DES` or `PBE-SHA1-3DES` cipher while generating the private key. Run one of the following commands:

```
openssl genrsa 2048 | openssl pkcs8 -topk8 -inform PEM -v1 PBE-SHA1-3DES -out rsa_key.p8
```

or:

```
openssl genrsa 2048 | openssl pkcs8 -topk8 -inform PEM -v1 PBE-SHA1-2DES -out rsa_key.p8
```

If you use a generic command without the `PBE-SHA1-2DES` or `PBE-SHA1-3DES` cipher, an error message about an invalid or unsupported private key might occur while fetching the target schema in the target definition step of the database ingestion task wizard.
The error message does not occur if you use OpenSSL 1.1.1 to generate the private key.

- Database ingestion incremental load jobs and combined initial and incremental load jobs generate a recovery table named INFORMATICA_CDC_RECOVERY on the target to store internal service information that prevents jobs restarted after a failure from propagating previously processed data again. This recovery table is generated in the same schema as the target tables.
- For Snowflake targets, you cannot alter the scale of NUMBER fields or change the data type of an existing field to a different data type because Snowflake does not support these actions.

Configuring connections

Configure a source connection and a target connection, if not already defined, from the **Connections** page in Administrator. Mass Ingestion Databases uses the Oracle Database Ingestion connector and Snowflake Data Cloud connector.

1. In Administrator, select **Connections**.
2. On the **Connections** page, click **New Connection**.
3. Configure the following connection details:

Property	Description
Connection Name	The name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + - Maximum length is 100 characters. Connection names are not case sensitive.
Description	A description of the connection.
Type	The type of connection, such as Snowflake Data Cloud.

After you select the connection type, additional properties that are specific to that type appear.

4. Configure the Oracle connection properties and the Snowflake connection properties.
5. To test each connection, click **Test Connection**.

   The results of the test are displayed at the top of the page.

   If a connection fails, contact the database administrator, or recheck your settings and verify that the selected runtime environment has the status of Up and Running.
6. Click **Save** to save the connection.

Oracle Database Ingestion connection properties

When you define an Oracle Database Ingestion connection for a database ingestion task, you must configure connection properties.

The following table describes the connection properties:

Property	Description
Connection Name	A name for the connection. This name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + - Spaces at the beginning or end of the name are trimmed and are not saved as part of the name. Maximum length is 100 characters. Connection names are not case sensitive.
Description	An optional description for the connection. Maximum length is 255 characters.
Type	The type of connection. For an Oracle Database Ingestion connection, the type must be **Oracle Database Ingestion**.
Runtime Environment	The name of the runtime environment where you want to run database ingestion tasks. You define runtime environments in Administrator.
User Name	User name for the Oracle database login. The user name cannot contain a semicolon.
Password	Password for the Oracle database login. The password cannot contain a semicolon.
Host	Host name of the database server.
Port	Network port number used to connect to the database server. Default is 1521.
Service Name	Service name or System ID (SID) that uniquely identifies the Oracle database. Specify the SID in the following format, including the leading semicolon (;), to connect to Oracle databases: `;SID=<ORACLE_SID>`
Schema	Schema used for the Oracle connection.
Code Page	The code page of the database server. Database ingestion tasks use the UTF-8 code page. Default is UTF-8.
Encryption Method	For initial load jobs, determines whether the data exchanged between the Secure Agent and the Oracle database server is encrypted: Options are: - **SSL**. Establishes a secure connection using SSL for data encryption. If the Oracle database server cannot configure SSL, the connection fails. - **No Encryption**. Establishes a connection without using SSL. Data is not encrypted. Default is No Encryption.

Property	Description
Crypto Protocol Version	If you selected SSL as the encryption method, you must specify a cryptographic protocol or a list of cryptographic protocols supported by your server to use with an encrypted connection. Options are: - SSLv2 - SSLv3 - TLSv1.2 Default is TLSv1.2.
Validate Server Certificate	If you selected SSL as the encryption method, controls whether the Secure Agent validates the server certificate that is sent by the Oracle database server. - **True**. Validate the server certificate. - **False**. Do not validate the server certificate. Default is False. If you also specify the **Host Name in Certificate** property, the Secure Agent also validates the host name in the certificate.
Trust Store	If you selected SSL as the encryption method and enabled validation of the server certificate, specify the path and name of the truststore file, which contains the list of the Certificate Authorities (CAs) that the client trusts for SSL authentication.
Trust Store Password	If you selected SSL as the encryption method and enabled validation of the server certificate, specify a password for accessing the contents of the truststore file.
Host Name in Certificate	If you selected SSL as the encryption method and enabled validation of the server certificate, specify the host name of the machine that hosts the Oracle database to provide for additional security. The Secure Agent validates the host name included the connection with the host name in the SSL certificate.
Key Store	If you selected SSL as the encryption method and client authentication is enabled on the Oracle database server, specify the path and name of the keystore file. The keystore file contains the certificates that the client sends to the Oracle server in response to the server's certificate request.
Key Store Password	If you selected SSL as the encryption method and client authentication is enabled on the Oracle database server, specify the password for the keystore file.
Key Password	If you selected SSL as the encryption method and client authentication is enabled on the Oracle database server, specify the password for the keys in the keystore file. Use this property when the keys have a different password than the keystore file.
Database Connect String	A TNS name, an Oracle Net keyword-value pair, or a SQL connect string URL that OCI uses to connect to Oracle.

Property	Description
TDE Wallet Directory	The path to the directory that contains the Oracle wallet file used for Oracle Transparent Data Encryption (TDE). Specify this property value only if you capture change data from TDE-encrypted tablespaces and one of the following conditions are true: - The Oracle wallet is not available to the database. - The Oracle database is running on a server that is remote from Oracle redo logs. - The wallet directory is not in the default location on the database host or the wallet name is not the default name of ewallet.p12. - The wallet directory is not available to the Secure Agent host.
TDE Wallet Password	A clear text password that is required to access the Oracle TDE wallet and get the master key. This property value is required if you need to read and decrypt data from TDE-encrypted tablespaces in the Oracle source database.
Directory Substitution	A local path prefix to substitute for the server path prefix of the redo logs on the Oracle server. This substitute local path is required when the log reader runs on a system other than the Oracle server and uses a different mapping to access the redo log files. Use this property in the following situations: - The redo logs reside on shared disk. - The redo logs have been copied to a system other than the Oracle system. - The archived redo logs are accessed by using a different NFS mount. **Note:** Do not use this statement if you use Oracle Automatic Storage Management (ASM) to manage the redo logs. You can define one or more substitutions. Use the following format: `server_path_prefix,local_path_prefix;server_path_prefix,local_path_prefix;...`
Reader Active Log Mask	A mask that the log reader uses for selecting active redo logs when the Oracle database uses multiplexing of redo logs. The log reader compares the mask against the member names in an active redo log group to determine which log to read. In the mask, you can use the asterisk (*) wildcard to represent zero or more characters. The mask can be up to 128 characters in length. It is case-sensitive on Linux or UNIX systems but not on Windows systems.
Reader Archive Destination 1	The primary log destination from which the log reader reads archived logs, when Oracle is configured to write more than one copy of each archived redo log. Enter a number that corresponds to a *n* value in an Oracle LOG_ARCHIVE_DEST_*n* initialization parameter, where *n* is a value from 1 to 10. If you set only one of the Reader Archive Destination 1 and Destination 2 properties, the log reader uses that property setting. If you specify neither property, the archive log queries are not filtered by the log destination.

Property	Description
Reader Archive Destination 2	The secondary log destination from which the log reader reads archived logs when the primary destination becomes unavailable or when the logs at the primary destination cannot be read. For example, logs might have been corrupted or deleted. Enter a number that corresponds to the *n* value in an Oracle LOG_ARCHIVE_DEST_*n* initialization parameter, where *n* is a value from 1 to 10. Usually, this value is a number greater than 1.
Reader ASM Connect String	In an Oracle ASM environment, the Oracle connection string, defined in TNS, that the log reader uses to connect to the ASM instance that manages storage of active and archived redo logs for the source database.
Reader ASM User Name	In an Oracle ASM environment, an Oracle user ID that the log reader uses to connect to the ASM instance that manages storage of active and archived redo logs for the source database. This user ID must have SYSDBA or SYSASM authority. To use SYSASM authority, set the **Reader ASM Connect As SYSASM** property to Y.
Reader ASM Password	In an Oracle ASM environment, a clear text password for the user that is specified in the **Reader ASM User Name** property. The log reader uses this password and the ASM user name to connect to the ASM instance that manages storage of active and archived redo logs for the source database.
Reader ASM Connect As SYSASM	If you use Oracle 11*g* ASM or later and want the log reader to use a user ID that has SYSASM authority to connect to the ASM instance, select this check box. Also specify a user ID that has SYSASM authority in the **Reader ASM User Name** property. To use a user ID that has SYSDBA authority, clear this check box. By default, this check box is cleared.

Property	Description
Reader Mode	Indicates the source of and types of Oracle redo logs that the log reader reads. Valid options are: - **ACTIVE**. Read active and archived redo logs from the Oracle online system. Optionally, you can use the **Reader Active Log Mask** property to filter the active redo logs and use the **Reader Archive Destination 1** and **Reader Archive Destination 2** properties to limit the archived log destinations from which to read archived logs. - **ARCHIVEONLY**. Read only archived redo logs. Optionally, you can use the **Reader Archive Destination 1** and **Reader Archive Destination 2** properties to limit the archived log destinations from which to read archived logs. - **ARCHIVECOPY**. Read archived redo logs that have been copied to an alternate file system. For combined initial and incremental load jobs, you must also set the source custom property pwx.cdcreader.oracle.reader.additional with the dir and file parameters, at the direction of Informatica Global Customer Support. You can use this option in the following situations: - You do not have the authority to access the Oracle archived redo logs directly. - The archived redo logs are written to ASM, but you do not have access to ASM. - The archived log retention policy for the database server causes the archived logs to not be retained long enough. With this option, the **Reader Archive Destination 1** and **Reader Archive Destination 2** properties are ignored. Default is **ACTIVE**.
Reader Standby Log Mask	A mask that the log reader uses for selecting redo logs for an Oracle physical standby database when the database uses multiplexing of redo logs. The log reader compares the mask against the member names in an redo log group to determine which log to read. In the mask, you can use the asterisk (*) wildcard to represent zero or more characters. The mask can be up to 128 characters in length. It is case-sensitive on Linux or UNIX systems but not on Windows systems.
Standby Connect String	An Oracle connection string, defined in TNS, that the log reader uses to connect to the Oracle physical standby database for change capture when the database is not open with read only access.
Standby User Name	A user ID that the log reader uses to connect to the Oracle physical standby database for change capture. This user ID must have SYSDBA authority.
Standby Password	A password that the log reader uses to connect to the Oracle physical standby database for change capture.

Property	Description
RAC Members	The maximum number of active redo log threads, or *members*, in an Oracle Real Application Cluster (RAC) that can be tracked. For a Data Guard physical standby database that supports a primary database in a RAC environment, this value is the number of active threads for the primary database. Valid values are 1 to 100. Default is 0, which causes an appropriate number of log threads to be determined automatically. If this value is not appropriate for your environment, set this property to a value greater than 0.
BFILE Access	Select this check box in the following circumstances: - You use BFILE access to redo logs in physical directories on the local Oracle server file system. BFILE access uses Oracle directory objects to remotely access the redo logs in the file system. This method is an alternative to other log access methods such as ASM or NFS mounts. - You have an Amazon Relational Database Service (RDS) for Oracle source. In this case, this option enables access to the redo logs of a cloud-based database instance deployed in RDS. By default, this check box is cleared.

Snowflake Data Cloud connection properties

When you set up a Snowflake Data Cloud connection, configure the connection properties.

You can use the following authentication methods to connect to Snowflake:

- Standard. Uses Snowflake account user name and password credentials to connect to Snowflake.

  Note: For application ingestion tasks, you can use only the Standard authentication method.
- Authorization Code. Uses the OAuth 2.0 protocol with Authorization Code grant type to connect to Snowflake. Authorization Code allows authorized access to Snowflake without sharing or storing your login credentials.
- KeyPair. Uses the private key file and private key file password, along with the existing Snowflake account user name to connect to Snowflake.
- Client Credentials. Uses the OAuth 2.0 protocol with the Client Credentials grant type to connect to Snowflake. Doesn't apply to Mass Ingestion application and database ingestion tasks.

You create a Snowflake Data Cloud connection on the Connections page. You can then use the connection when you read from or write data to Snowflake.

Standard authentication

When you set up a Snowflake Data Cloud connection, configure the connection properties.

The following table describes the Snowflake Data Cloud connection properties for the Standard authentication mode:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Type	The Snowflake Data Cloud connection type.
Runtime Environment	The name of the runtime environment where you want to run the tasks. You cannot run application ingestion tasks and database ingestion tasks on a Hosted Agent or serverless runtime environment.
Authentication	The authentication method that the connector must use to log in to Snowflake. Select **Standard**. Default is **Standard**.
Username	The user name to connect to the Snowflake account.
Password	The password to connect to the Snowflake account.
Account	The name of the Snowflake account. For example, if the Snowflake URL is https://<123abc>.us-east-2.aws.snowflakecomputing.com/console/login#/, your account name is the first segment in the URL before snowflakecomputing.com. Here, 123abc.us-east-2.aws is your account name. If you use the Snowsight URL, for example, https://app.snowflake.com/us-east-2.aws/<123abc>/dashboard, your account name is 123abc.us-east-2.aws **Note:** Ensure that the account name doesn't contain underscores. To use an alias name, contact Snowflake Customer Support.
Warehouse	The Snowflake warehouse name.
Role	The Snowflake role assigned to the user.

Additional JDBC URL Parameters	The additional JDBC connection parameters.
	Enter one or more JDBC connection parameters in the following format:
	`<param1>=<value>&<param2>=<value>&<param3>=<value>....`
	For example, pass the database and schema values when you connect to Snowflake:
	`db=mydb&schema=public`
	Important: Ensure that there is no space before and after the equal sign (=) when you add the parameters.

OAuth 2.0 authorization code authentication

The following table describes the Snowflake Data Cloud connection properties for an OAuth 2.0 - AuthorizationCode type connection:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Type	The Snowflake Data Cloud connection type.
Runtime Environment	The name of the runtime environment where you want to run the tasks. You cannot run application ingestion tasks and database ingestion tasks on a Hosted Agent or serverless runtime environment.
Authentication	The authentication method that Snowflake Data Cloud Connector must use to log in to Snowflake. Select **AuthorizationCode**.
Account	The name of the Snowflake account. For example, if the Snowflake URL is https://<123abc>.us-east-2.aws.snowflakecomputing.com/console/login#/, your account name is the first segment in the URL before snowflakecomputing.com. Here, 123abc.us-east-2.aws is your account name. If you use the Snowsight URL, for example, https://app.snowflake.com/us-east-2.aws/<123abc>/dashboard, your account name is 123abc.us-east-2.aws **Note:** Ensure that the account name doesn't contain underscores. To use an alias name, contact Snowflake Customer Support.
Warehouse	The Snowflake warehouse name.

Additional JDBC URL Parameters	The additional JDBC connection parameters.
	Enter one or more JDBC connection parameters in the following format:
	`<param1>=<value>&<param2>=<value>&<param3>=<value>....`
	For example, pass the database and schema values when you connect to Snowflake:
	`db=mydb&schema=public`
	Important: Ensure that there is no space before and after the equal sign (=) when you add the parameters.

Authorization URL	The Snowflake server endpoint that is used to authorize the user request. The authorization URL is https://<account name>.snowflakecomputing.com/oauth/authorize, where <account name> specifies the full name of your account provided by Snowflake. For example, https://<abc>.snowflakecomputing.com/oauth/authorize **Note:** If the account name contains underscores, use the alias name. You can also use the Authorization Code grant type that supports the authorization server in a Virtual Private Cloud network.

Property	Description
Access Token URL	The Snowflake access token endpoint that is used to exchange the authorization code for an access token. The access token URL is https://<account name>.snowflakecomputing.com/oauth/token-request, where <account name> specifies the full name of your account provided by Snowflake. For example, https://<abc>.snowflakecomputing.com/oauth/token-request **Note:** If the account name contains underscores, use the alias name.
Client ID	Client ID of your application generated when you create a security integration of type OAuth in Snowflake. For more information, see the Snowflake documentation. Not used by Mass Ingestion application and database ingestion tasks.
Client Secret	Client secret generated for the client ID. Not used by Mass Ingestion application and database ingestion tasks.
Scope	Determines the access control if the API endpoint has defined custom scopes. Enter space-separated scope attributes. For example, specify `session:role:CQA_GCP` as the scope to override the value of the default user role. The value must be one of the roles assigned in Security Integration. Not used by Mass Ingestion application and database ingestion tasks.
Access Token Parameters	Additional parameters to use with the access token URL. Define the parameters in the JSON format. For example, define the following parameters: `[{"Name":"code_verifier","Value":"5PMddu6Zcg6Tc4sbg"}]` Not used by Mass Ingestion application and database ingestion tasks.

Authorization Code Parameters	Additional parameters to use with the authorization token URL.
	Define the parameters in the JSON format.
	For example, define the following parameters:
	`[{"Name":"code_challenge","Value":"Ikr-vv52th0UeVRi4"},`
	`{"Name":"code_challenge_method","Value":"S256"}]`
	Not used by Mass Ingestion application and database ingestion tasks.

Access Token	The access token value. Enter the populated access token value, or click **Generate Token** to populate the access token value.
Generate Token	Generates the access token and refresh token based on the OAuth attributes you specified.
Refresh Token	The refresh token value. Enter the populated refresh token value, or click **Generate Token** to populate the refresh token value. If the access token is not valid or expires, the agent fetches a new access token with the help of the refresh token. **Note:** If the refresh token expires, provide a valid refresh token or regenerate a new refresh token by clicking **Generate Token**. Not used by Mass Ingestion application and database ingestion tasks.

Key pair authentication

The following table describes the Snowflake Data Cloud connection properties for the KeyPair authentication type connection:

Connection property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Type	The Snowflake Data Cloud connection type.
Runtime Environment	The name of the runtime environment where you want to run the tasks. You cannot run application ingestion tasks and database ingestion tasks on a Hosted Agent or serverless runtime environment.
Authentication	The authentication method to log in to Snowflake. Select **KeyPair**.
Username	The user name to connect to the Snowflake account.
Account	The name of the Snowflake account. For example, if the Snowflake URL is https://<123abc>.us-east-2.aws.snowflakecomputing.com/console/login#/, your account name is the first segment in the URL before snowflakecomputing.com. Here, 123abc.us-east-2.aws is your account name. If you use the Snowsight URL, for example, https://app.snowflake.com/us-east-2.aws/<123abc>/dashboard, your account name is 123abc.us-east-2.aws. **Note:** Ensure that the account name doesn't contain underscores. To use an alias name, contact Snowflake Customer Support.
Warehouse	The Snowflake warehouse name.

Additional JDBC URL Parameters	The additional JDBC connection parameters.
	Enter one or more JDBC connection parameters in the following format:
	`<param1>=<value>&<param2>=<value>&<param3>=<value>....`
	For example, pass the database and schema values when you connect to Snowflake:
	`db=mydb&schema=public`
	Important: Ensure that there is no space before and after the equal sign (=) when you add the parameters.

Private Key File	Path to the private key file, including the private key file name, that the Secure Agent uses to access Snowflake. For example, specify the following path and key file name: - On Windows: `C:\Users\path_to_key_file\rsa_key.p8` - On Linux: `/export/home/user/path_to_key_file/rsa_key.p8` **Note:** Verify that the keystore is FIPS-certified.
Private Key Password	Password for the private key file.

Configuring a database ingestion task

In Mass Ingestion, use the database ingestion task wizard to configure a database ingestion task.

On the wizard pages, complete the following configuration tasks:

1. Define basic task information, such as the task name, project location, runtime environment, and load type.
2. Configure the source.
3. Configure the target.
4. Configure runtime options.

Click **Next** or **Back** to navigate from one page to another. At any point, you can click **Save** to save the information that you have entered so far.

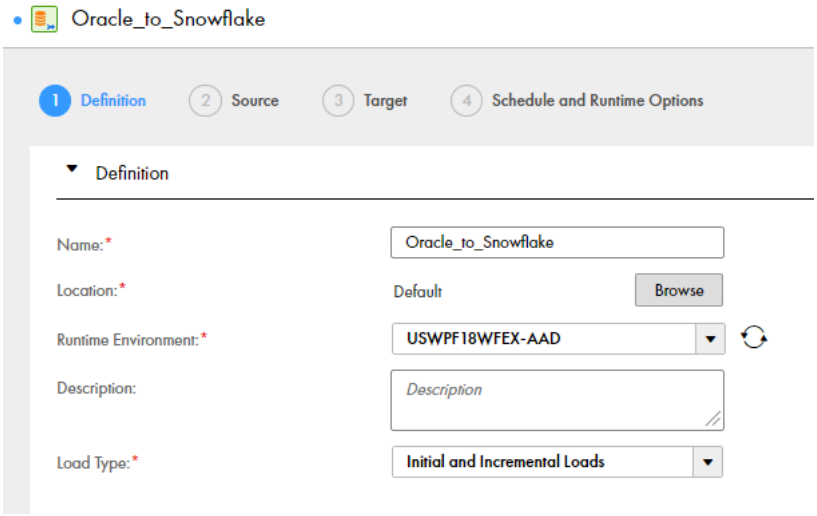
After you complete all wizard pages, save the information and then click **Deploy** to make the task available as an executable job to the Secure Agent.

Step 1: Defining basic task information

To begin defining a database ingestion task, you must first enter some basic information about the task, such as a task name, project or project folder location, and load operation type.

1. In Mass Ingestion, click **New** > **Database Ingestion Task**.

   The **Definition** page of the Mass Ingestion Databases Task wizard appears.



2. Configure the following properties:

Property	Description
Name	A name for the database ingestion task. Task names can contain Latin alphanumeric characters, spaces, periods (.), commas (,), underscores (_), plus signs (+), and hyphens (-). Task names cannot include other special characters. Task names are not case sensitive. Maximum length is 50 characters. **Note:** If you include spaces in the database ingestion task name, after you deploy the task, the spaces do not appear in the corresponding job name.
Location	The project or project\folder that will contain the task definition. The default is the currently selected project or project subfolder in Explore. If a project or project subfolder is not selected, the default is the **Default** project.
Runtime Environment	The runtime environment in which you want to run the task. The runtime environment must be a Secure Agent group that consists of one or more Secure Agents. A Secure Agent is a lightweight program that runs tasks and enables secure communication. You cannot use the Hosted Agent or a serverless runtime environment. **Tip:** Click the **Refresh** icon to refresh the list of runtime environments.
Description	An optional description for the task. Maximum length is 4,000 characters.
Load Type	The type of load operation that the database ingestion task performs. For this scenario, select **Initial and Incremental Loads** to perform an initial load of point-in-time data to the target and then automatically switch to propagating incremental data changes made to the same source tables on a continuous basis. **Note:** If a change record is captured during the initial unload load phase, it's withheld from apply processing until after the unload phase completes. Any insert rows captured during the unload phase are converted into a pair of delete and insert operations so that only one insert row is applied to the target in the case where the insert occurs in both the unloaded data and the captured change data.

3. Click **Next**.

Step 2: Configuring the Oracle source

Configure the Oracle source on the **Source** page of the database ingestion task wizard.

1. In the **Connection** list, select the connection for the Oracle source that you configured and tested in Administrator for a runtime environment that your organization uses. The connection type appears in parentheses after the connection name.

   The list includes only the connection types that are valid for the load type selected on the **Definition** page. No connections are listed if you did not select a load type.

   Note: After you deploy the ingestion task, you cannot change the connection without first undeploying the associated ingestion job. After you change the connection, you must deploy the task again.

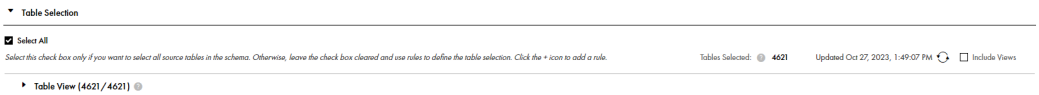
2. In the **Schema** list, select the source schema that includes the source tables.

   When creating a task that has a Oracle source, the schema name that is specified in the connection properties is displayed by default.

3. Under **Change Data Capture Method**, set the **CDC Method** field to **Log-based CDC**.

In this CDC method, Inserts, Updates, Deletes, and column DDL changes are captured from the Oracle redo logs.

4. Under **Table Selection**, select **Select All** to select all tables and columns in the schema for data replication. The **Tables Selected** field shows the count of all selected tables. You won't be able to edit the selection later under **Table View**.



Note: Fetching information for all of the tables might take a long time.

Alternatively, you can use rules to define a subset of source tables to replicate.

5. To use rules to select the source tables, make sure that the **Select All** check box is cleared and then add rules.

When rule-based selection is used, you can refine the set of selected tables by table under **Table View** and also set an option for trimming spaces in character data.

Note: The default "Include *" rule selects all tables in the specified schema. To see how many tables are selected by this rule, click the Refresh icon to display the table count in **Total Tables Selected** and click **Apply Rules** to see the table count in **Table View**.

To add a rule:

a. Under **Rules**, click the Add Rule (+) icon above the first table. A row is added to the table.

b. In the **Table Rule** column, select **Include** or **Exclude** to create an inclusion or exclusion rule, respectively.

c. In the **Condition** column, enter a table name or a table-name mask that includes one or more wildcards to identify the source tables to include in or exclude from table selection. Use the following guidelines:

- A mask can contain one or both of the following wildcards: an asterisk (*) wildcard to represent one or more characters and a question mark (?) wildcard to represent a single character. A wildcard can occur multiple times in a mask value and can occur anywhere in the value.
- The task wizard is case sensitive. Enter the table names or masks in the case with which the tables were defined.
- Do not include delimiters such as quotation marks or brackets, even if the source database uses them. For example, Oracle requires quotation marks around lowercase and mixed-case names to force the names to be stored in lowercase or mixed case. However, in the task wizard, you must enter the lowercase or mixed-case names without quotation marks.
- If a table name includes special characters such as a backslash (\), asterisk(*), dollar sign ($), caret (^), or question mark (?), escape each special character with a backslash (\) when you enter the rule.

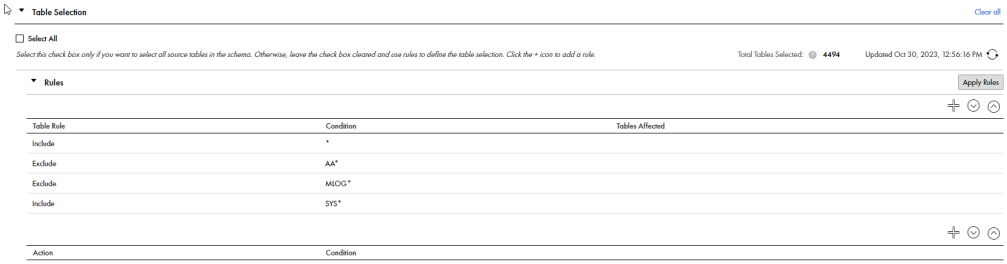
d. Define additional rules as needed.

The rules will be processed in the order in which they're listed, from top to bottom. Use the arrow icons to change the order.

e. When finished, click **Apply Rules**.

The **Total Tables Selected** and **Tables Affected** counts are updated.

The following image shows multiple rules defined on the **Source** page:



After you apply rules, if you add, delete, or change rules, you must click **Apply Rules** again. Click the Refresh icon to update the table counts. If you delete all rules without clicking **Apply Rules**, a validation error will occur at deployment, even if the **Table View** list still lists tables. If you switch to **Select All**, the rules are no longer in effect and disappear.

6. To perform trim actions on character columns in the source tables selected based on rules, create column action rules.

   To create a column action rule:

   a. Click the Add Rule (+) icon above the second table.

   b. In the **Action** column, select one of the following options:

      - **LTRIM**. Trims spaces to the left of character column values.
      - **RTRIM**. Trims spaces to the right of character column values.
      - **TRIM**. Trims spaces to the left of and to the right of character column values.

   c. In the **Condition** column, enter a column name or a column name mask that includes one or more asterisk (*) or question mark (?) wildcards. The value is matched against columns in the selected source tables to identify the columns to which the action applies.

   Note: You can define multiple rules for different action types or for the same action type with different conditions. The rules are processed in the order in which they're listed, from top to bottom. Use the arrow icons to change the order.

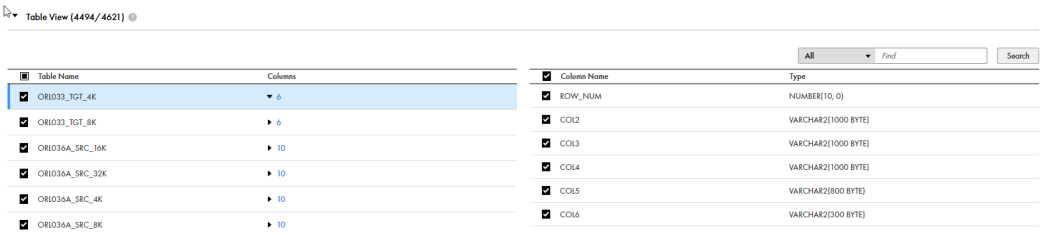
7. Under **Table View**, view or edit the set of selected source tables and columns.

   If you selected **Select All**, the lists of tables and columns are view only.

   If you applied rules, you can refine the set of selected tables by clicking the check box next to individual tables. Deselect any tables that you do not want to replicate, or select additional items to replicate. Click the Refresh icon to update the selected tables count.

   You can also individually deselect or reselect the columns in a selected source table. To view or change the columns from which data will be replicated for a selected table, click the highlighted number of columns in the **Columns** column. The column names and data types are displayed to the right. By default, all columns are selected for a selected source table. To deselect a column or reselect it, click the check box next to the column name. You cannot deselect a primary key column.

   The following image shows selected tables and the selected columns for the first table:



Notes:

- To search for tables or columns or both, in the drop-down list above the columns list, select **Table Name**, **Columns**, or **All** and then enter a search string in the *Find* box and click **Search**. You can include a single asterisk (*) wildcard at the beginning or end of the string.
- The first time you change a check box setting for a table or column, the rules are no longer in effect. The selections under **Table View** take precedence. However, if you click the Add Rule (+) icon again, any tables that you deselected or selected individually are reflected as new rules in the Rules list and the rules once again take precedence. If you want to return to the **Table View** list, click **Apply Rules** again.
- In the **Table View** section, the length of character columns is shown in bytes. Since the number of bytes per character differs depending on the character set encoding used by the database, for some sources, the actual column length in number of characters might be different than the number of bytes shown in the **Table View**.
- If you select columns individually, the resulting set of columns is fixed and will not be updated by any schema change, regardless of schema drift settings. For example, if a source column is added or renamed, that column is silently excluded from CDC processing because it's not in the list of selected columns. However, if a selected column is dropped on the source, the schema drift Drop Column option controls how it's handled. The dropped column operation is not reflected in the list of columns until you apply rules again.

8. If one or more of the selected source tables are not enabled for change data capture, you can generate a script for enabling CDC and then run or download the script.
   a. In the **CDC Script** field, select one of the following options:
      - **Enable CDC for all columns**. Enables CDC for all columns in the selected source tables.

        Note: For source tables without a primary key, including any tables with unique indexes, CDC is enabled for all columns by default, regardless of which option is selected.
      - **Enable CDC for primary key columns**. Enables CDC only for primary key columns in the selected source tables.

      The script enables supplemental logging for all or primary key columns in the selected source tables to log additional information in the redo logs.

   b. To run the script, click **Execute**.

      If you do not have a database role or privilege that allows you to run the script, click the Download icon to download the script. The script file name has the following format: `cdc_script_taskname_number.txt`. Then ask your database administrator to run the script.

      Make sure the script runs before you run the database ingestion task.

Note: If you change to the **CDC Script** option later and run the script again, the script first drops CDC for the original set of columns and then enables CDC for the current set of columns.

9. To create and download a list of the source tables that match the table selection criteria, perform the following substeps:
   a. If you used rule-based table selection, in the **List Tables by Rule Type** list, select the type of selection rules that you want to use. Options are:
      - **Include Rules Only**
      - **Exclude Rules Only**
      - **Include And Exclude Rules**
   b. To include columns in the list, regardless of which table selection method you used, select the **Include Columns** check box.
   c. Click the Download icon.

      A downloaded list that includes columns has the following format:

      ```
      status,schema_name,table_name,object_type,column_name,comment
      ```

      The following table describes the information that is displayed in the downloaded list:

Field	Description
status	Indicates whether Mass Ingestion Databases excludes the source table or column from processing because it has an unsupported type. Valid values are: - **E**. The object is excluded from processing by an Exclude rule. - **I**. The object is included in processing. - **X**. The object is excluded from processing because it is an unsupported type of object. For example, unsupported types of objects include columns with unsupported data types and tables that include only unsupported columns. The comment field provides detail on unsupported types.
schema_name	Specifies the name of the source schema.
table_name	Specifies the name of the source table.
object_type	Specifies the type of the source object. Valid values are: - **C**. Column. - **T**. Table.
column_name	Specifies the name of the source column. This information appears only if you selected the **Columns** check box.
comment	Specifies the reason why a source object of an unsupported type is excluded from processing even though it matches the selection rules.

10. Under **Advanced**, set the advanced properties that are available for combined initial and incremental load jobs with Oracle sources.

Property	Description
Enable Persistent Storage	Select this check box to enable persistent storage of transaction data in a disk buffer so that the data can be consumed continually, even when the writing of data to the target is slow or delayed. Benefits of using persistent storage are faster consumption of the source transaction logs, less reliance on log archives or backups, and the ability to still access the data persisted in disk storage after restarting a database ingestion job.
Enable Partitioning	Select this check box to enable partitioning of source objects. When an object is partitioned, the database ingestion job processes the records read from each partition in parallel. For Oracle sources, Mass Ingestion Databases determines the range of partitions by using the ROWID as the partition key. **Note:** In combined initial and incremental loads, the partitioning of source objects occurs only in the initial load phase.
Number of Partitions	If you enable partitioning of source objects, enter the number of partitions you want to create. The default number is 5. The minimum value is 2.
Initial Start Point for Incremental Load	Set this field if you want to customize the position in the source logs from which the database ingestion job starts reading change records the first time it runs. Options are: - **Latest Available**. The latest available position in the database log or structure. - **Position**. A valid Oracle SCN that is used to determine a position in the change stream from which to start retrieving change records. This value must be equal to or less than the current position value. An invalid value will cause the job to fail. The default value is 0, which causes processing to start from the latest available position. - **Specific Date and Time**. A date and time, in the format MM/DD/YYYY hh:mm AM|PM, that Mass Ingestion Databases uses to determine the position in the change stream from which to start retrieving change records. Mass Ingestion Databases retrieves only the changes that were started after this date and time. If you enter a date and time earlier than the earliest date and time in the available archived logs, the job will fail. The default is **Latest Available**. **Notes:** - The **Initial Start Point for Incremental Load** option pertains only to the initial run of a job. Thereafter, if you resume a stopped or aborted job, the job begins propagating source data from where it last left off. - For combined initial and incremental load jobs, initial loading is not performed until the incremental processing of change data reaches the end of the current transaction log. For this reason, Informatica recommends that you select **Latest Available** as the start point. For Oracle combined initial and incremental load jobs, Oracle Flashback queries are used to get committed data that was current at a specific point in the change stream. Ensure that no source table is truncated during the initial load period. If truncation occurs, any DDL change performed during a flashback query causes the query to fail.

11. Under **Custom Properties**, you can specify custom properties at the direction of Informatica Global Customer Support to meet your special requirements. In this scenario, skip this step.
12. Click **Next**.

Step 3: Configuring the Snowflake target

Configure the target on the **Target** page of the database ingestion task wizard.

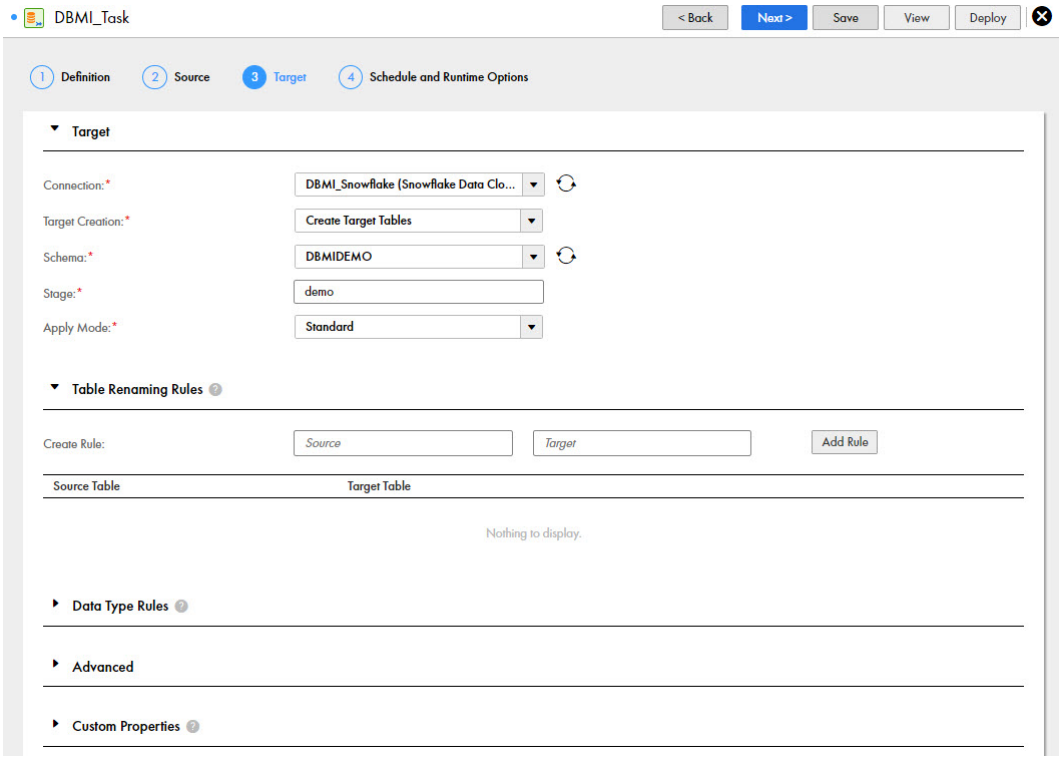
1. In the **Connection** list, select the Snowflake connection that you previously defined and tested in Administrator for the runtime environment. The connection type appears in parentheses after the connection name.

   The list includes only the connection types that are valid for the load type selected on the **Definition** page. No connections are listed if you did not select a load type.

   If you change the load type and the selected connection is no longer valid, a warning message is issued and the **Connection** field is cleared. You must select another connection that is valid for the updated load type.

   Note: After you deploy the ingestion task, you cannot change the connection without first undeploying the associated ingestion job. You must then deploy the task again.

2. In the **Target** section, configure the properties that pertain to a Snowflake target.



The following table describes these properties:

Property	Description
Target Creation	The only available option is **Create Target Tables**, which generates the target tables based on the source tables. **Note:** After the target table is created, Mass Ingestion Databases intelligently handles the target tables on subsequent job runs. Mass Ingestion Databases might truncate or re-create the target tables depending on the specific circumstances.
Schema	Select the target schema in which Mass Ingestion Databases creates the target tables.

Property	Description
Stage	The name of internal staging area that holds the data read from the source before the data is written to the target tables. This name must not include spaces. If the staging area does not exist, it will be automatically created.
Apply Mode	Indicates how source DML changes, including inserts, updates, and deletes, are applied to the target. Options are: - **Standard**. Accumulate the changes in a single apply cycle and intelligently merge them into fewer SQL statements before applying them to the target. For example, if an update followed by a delete occurs on the source row, no row is applied to the target. If multiple updates occur on the same column or field, only the last update is applied to the target. If multiple updates occur on different columns or fields, the updates are merged into a single update record before being applied to the target. - **Soft Deletes**. Apply source delete operations to the target as soft deletes. A soft delete marks the deleted row as deleted without actually removing it from the database. For example, a delete on the source results in a change record on the target with "D" displayed in the INFA_OPERATION_TYPE column. If an update followed by a delete occurs on the source, two records are written to the target both with "D" displayed in the INFA_OPERATION_TYPE column. Consider using soft deletes if you have a long-running business process that needs the soft-deleted data to finish processing, to restore data after an accidental delete operation, or to track deleted values for audit purposes. **Note:** If you use **Soft Deletes** mode, you must not perform an update on the primary key in a source table. Otherwise, data corruption can occur on the target. - **Audit**. Apply an audit trail of every DML operation made on the source tables to the target. A row for each DML change on a source table is written to the generated target table along with the audit columns you select under the **Advanced** section. The audit columns contain metadata about the change, such as the DML operation type, time, owner, transaction ID, generated ascending sequence number, and before image. Consider using Audit apply mode when you want to use the audit history to perform downstream computations or processing on the data before writing it to the target database or when you want to examine metadata about the captured changes. The default value is **Standard**. ***In this scenario, accept the default value of **Standard**.***

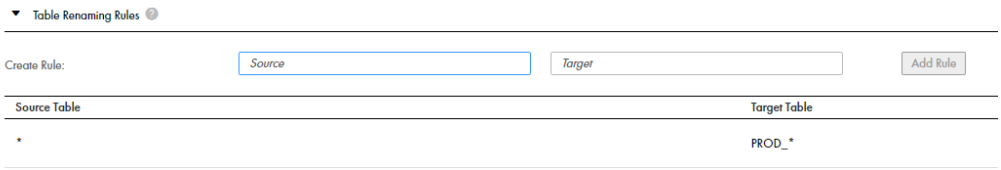
3. If you want to rename the target objects that are associated with the selected source tables, define table renaming rules. For example, you might want to add a prefix or suffix. To create a rule for renaming tables:
   a. Under **Table Renaming Rules**, in the **Create Rule** fields, enter a source table name or a table name mask that includes one or more wildcards. Then enter the corresponding target table name or table name mask.

      Notes:

      - For the source, you can enter only the asterisk (*) wildcard to select all source tables that match the table selection criteria on the **Source** page. Or you can enter a specific source table name or a table-name mask that includes one or more of the following wildcards: an asterisk (*) to represent one or more characters or a question mark (?) to represent a single character.
      - To use a table-name mask with the wildcard character for the target, you must also use the wildcard character in the source. If you use a specific source table name with a target table mask that includes the wildcard character, the task deployment will fail.
      - If a table name includes special characters, such as a backslash (\), asterisk(*), dot (.), or question mark (?), escape each special character in the name with a backslash (\).

b. Click **Add Rule**

The rule appears in the rules list.



You can define multiple table rules. The order of the rules does not matter with regard to how they are processed unless a table matches multiple rules. In this case, the last matching rule determines the name of the table.

4. If you want to override the default mappings of source data types to target data types, define data type rules.
5. Under **Advanced**, you can enter the following advanced target properties which will occur in target tables:

Field	Description
Add Last Replicated Time	Select this check box to add a metadata column that records the timestamp at which a record was inserted or last updated in the target table. For initial loads, all loaded records have the same timestamp. For incremental and combined initial and incremental loads, the column records the timestamp of the last DML operation that was applied to the target. By default, this check box is not selected.
Prefix for Metadata Columns	Add a prefix to the names of the added metadata columns to easily identify them and to prevent conflicts with the names of existing columns. The default value is INFA_.
Superpipe	Snowpipe Streaming API allows you to quickly stream rows of data directly to Snowflake Data Cloud target tables with low latency instead of first writing the data to stage files. By default, this check box is selected. ***In this scenario, **deselect** the check box.***
Merge Frequency	When **Superpipe** is selected, you can optionally set the frequency, in seconds, at which change data rows are merged and applied to the Snowflake target tables.

6. Under **Custom Properties**, you can specify custom properties at the direction of Informatica Global Customer Support to meet your special requirements. In this scenario, skip this step.
7. Click **Next**.

Step 4: Configuring runtime options

On the **Schedule and Runtime Options** page of the database ingestion task wizard, you can configure some runtime options.

1. Under **Schema Drift Options**, specify the schema drift option to use for each of the supported types of DDL operations.

   The types of supported DDL operations are:

   - Add Column
   - Modify Column

- Drop Column
- Rename Column

The following table describes the schema drift options that you can set for a DDL operation type:

Option	Description
Ignore	Do not replicate DDL changes that occur on the source database to the target. For Snowflake targets, this option is the default option for the Drop Column and Rename Column operation types
Replicate	Replicate the DDL operation to the target. For Snowflake targets, this option is the default option for the Add Column and Modify Column operation types. **Restrictions:** - If you try to replicate a type of schema change that is not supported on the target, database ingestion jobs associated with the task will end with an error. - Add Column operations that add a primary-key column are not supported and can cause unpredictable results. - Modify Column operations that change the NULL or NOT NULL constraint for a column are not replicated to the target by design because changing the nullibility of a target column can cause problems when subsequent changes are applied. - DDL operations are replicated to the target only after some data changes have been captured.
Stop Job	Stop the entire database ingestion job.
Stop Table	Stop processing the source table on which the DDL change occurred. When one or more of the tables are excluded from replication because of the **Stop Table** schema drift option, the job state changes to **Running with Warning**. **Important:** The database ingestion job cannot retrieve the data changes that occurred on the source table after the job stopped processing it. Consequently, data loss might occur on the target. To avoid data loss, you will need to resynchronize the source and target objects that the job stopped processing. Use the **Resume With Options** > **Resync** option.

2. Under **Advanced**, optionally edit the **Number of Rows in Output File** value to specify the maximum number of rows that the database ingestion task writes to an output data file for the Snowflake target.

   Change data is flushed to the target either when this number of rows is reached or when the flush latency period expires and the job is *not* in the middle of processing a transaction. The flush latency period is the time that the job waits for more change data before flushing data to the target. The latency period is internally set to 10 seconds and cannot be changed.

   Valid values are 1 through 100000000. The default value is 100000 rows.

   Note: For Snowflake targets, the data is first stored in an internal stage area before being written to the target tables.

3. Under **Schedule**, the scheduling options are unavailable for incremental load tasks or combined initial and incremental load. Skip this field.
4. Under **Custom Properties**, you can specify custom properties that Informatica provides to meet your special requirements. Specify these properties only at the direction of Informatica Global Customer Support.
5. Click **Save**.

Deploying the database ingestion task

After you define a database ingestion task and save it, deploy the task to create an executable job instance on the on-premises system that contains the Secure Agent and the Database Ingestion agent service and DBMI packages. You must deploy the task before you can run the job. The deploy process also validates the task definition.

Before you deploy a task that has a Snowflake target, drop any existing target tables that do not match the structure of the source tables, for example, because of added or dropped source columns or altered column null constraints or data types. When you deploy the task, the target tables are generated based the latest source structure.

1. In the database ingestion task wizard, save the completed task definition.
2. To deploy a task, click **Deploy**.

After you deploy a task successfully, the associated job instance has the status of Deployed. You can run the job from either the **My Jobs** page in Mass Ingestion or from the **All Jobs** tab on the **Mass Ingestion** page in Operational Insights.

Deployment considerations:

- If you included spaces in the database ingestion task name, the spaces are omitted from the corresponding job name for the generated job instance.
- If the deployment process fails, the job status switches to Failed. You can then Undeploy the job. To diagnose the error, download the error log from the **My Jobs** page in Mass Ingestion or from the **All Jobs** tab on the **Mass Ingestion** page in Operational Insights. In the Actions menu for the job, click **Error Log**. After you resolve the problem, deploy the task again from the database ingestion task wizard.
- If you undeploy a job and then want to run a job for the associated ingestion task again, you must deploy the task again to create a new job instance. The new job instance name ends with an incremented number in the format *taskname-job_instance_number*. The job instance number is incremented each time you deploy the ingestion task by adding 1 to the maximum instance number across all ingestion jobs.
- If the Secure Agent is restarted while the task is deploying, the job status switches to Failed. Avoid restarting the Secure Agent while tasks are being deployed.
- If a task appears to be hung in the Deploying state, restart the Secure Agent. The associated job instance acquires the status of Failed. You can undeploy it and then deploy it again.

Running the deployed database ingestion job

You can run a database ingestion job after it has been deployed and is in a state other than Undeployed from either the **My Jobs** page in the Mass Ingestion service or from the **All Jobs** tab on the **Mass Ingestion** page in Operational Insights.

1. Navigate to the row for the job that you want to run.
2. In the **Actions** menu at the right end of the row, click **Run**.

   A subtask is started for each source table.

Monitoring database ingestion jobs

You can monitor the progress, performance, and status of ingestion jobs from the Mass Ingestion and Operational Insights services.

Depending on the service you use and type of ingestion job, you can view following monitoring information:

- On the **My Jobs** page in the Mass Ingestion service, monitor the ingestion jobs for the ingestion tasks that you deployed. You can view a list of your jobs that includes general job properties such as the task type, runtime environment, start time, duration, and current job status.

- On the **Mass Ingestion** page in Operational Insights service, monitor *all* types of ingestion jobs that any member of your organization deployed. You can view the following types of information:
  - Summary counts of ingestion jobs by task type and job status.
  - Recent jobs that require your attention because they failed or are running with errors or warnings.
  - A list of all ingestion jobs by type, including the general job properties.
- From either the list of your jobs or the list of all jobs, you can drill down to details for a specific job by clicking the job name. You can view overview job information, source object processing details, and alerts.

Note: Usually, a job name corresponds to the ingestion task name. For database ingestion jobs, the job name has the format *taskname-job_instance_number*. The number is incremented each time the job is deployed.

Other resources

For more information, visit the Informatica Documentation Portal .

You can also access the latest Mass Ingestion and Administrator help information from within the Informatica Intelligent Cloud Services interface.

Author

Informatica Documentation Team